

SEED STENCILS FOR DIFFUSION THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. A diffusion coefficient can vary smoothly on the scale of the grid, jump across an interface, or change smoothly over a width δ smaller than the node spacing h . Through such a sub-grid edge, standard high-order stencils that sample the coefficient at the nodes are first order until the grid resolves the edge, with a knee at $h \approx \delta$, and the interface-aware stencils built for a jump, applied at the edge's centre, sit on an error floor of order δ and leave it once the grid resolves the edge. We keep the nodes and replace the polynomial basis of the stencils that see the edge by *seeds*. The seeds are the profiles of solutions polynomial in time, continued through the edge by ordinary differential equations along the normal or along the edge's own coordinates; in one dimension they are the classical formal powers of the Sturm–Liouville operator, and as $\delta \rightarrow 0$ they reduce to the interface-aware stencils of a jump. On scattered nodes the resulting RBF-FD stencils are fourth order at every δ tested, elliptic and parabolic, with no switch between regimes. We compare them with the coefficient treatments of diffusion practice. The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at $\delta = 0$, ahead of the seeds below 801 nodes, where the seeds' line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

1. INTRODUCTION

Heat conduction through a composite material is governed by

$$(1) \quad u_t = \nabla \cdot (\alpha \nabla u) \equiv Lu$$

and, at equilibrium, by $Lu = 0$, with a conductivity $\alpha > 0$ that changes sharply between materials. At the two ends of the range of media the numerical picture is settled. Where α varies smoothly on the scale of the grid, a high-order stencil samples it at the nodes and keeps its order. Where α

jumps across an interface, the temperature and the flux $\alpha \partial_n u$ are continuous while the gradient is not, a standard stencil across the interface is first order, and the remedies rebuild the stencils or the elements that the interface cuts so that they satisfy the interface conditions (Section 1.1). One of them, the interface-aware RBF-FD stencils of [Mar16, MF17b] that this paper extends, keeps fourth order on scattered nodes, elliptic and parabolic, with a few rows of nodes placed across each interface. Between the two ends lies a third case: an edge that is smooth, so that no interface condition applies, but changes over a width δ smaller than the node spacing h . Every material edge is smooth on some scale, a model that resolves an edge on a fine grid presents it as sub-grid on a coarser one, and any regularisation of a jump produces one.

A standard stencil through a sub-grid edge samples α on its two sides, sees a jump, and treats the coefficient as smooth between its nodes, which is the first-order failure of the jump case. It recovers once the grid resolves the edge. In the one-dimensional ramp problem of Section 4.2, the error of the standard fourth-order operator through a 9 : 1 edge of width $\delta = 0.01$ or 0.0025 falls by 100 to 200 times between $h = 2\delta$ and $h = \delta/2$, where a first-order line falls by four. On scattered nodes in two dimensions (Section 6.2) its elliptic error stays at 0.86 to 0.94 of its error at the jump while $h \gtrsim 6\delta$, falls to 0.081 to 0.084 of it at $h \approx \delta$ for the three widths that reach it, and converges at the fifth order of the bulk stencils below. We call the turn at $h \approx \delta$ the knee.

The interface-aware stencils do not reach this regime either. Applied with the centre of the edge as the interface and the two sides' coefficients as if δ were 0, they converge to the jump problem's solution, which is not the smooth problem's: the resistance $\int dx/\alpha$ of a tanh edge between conductivities a and b falls short of the jump's by $c\delta$, $c = (a - b) \ln(a/b)/(2ab)$, so their error sits on a floor of order δ while the grid does not resolve the edge. Once it does, the rebuilt rows keep enforcing a kink that the solution lacks, and their error grows: through a 9 : 1 edge with $\delta = 0.01$ it reaches 0.11 at 1600 nodes in the one-dimensional equilibrium of Section 4.3. On scattered nodes they beat the standard stencils only while $h \gtrsim 3\delta$, and by at most a factor of 6. Using them means knowing δ and switching them off.

This paper keeps everything about the scheme except the basis of the stencils that see the edge. In place of the monomials a stencil is made exact on *seeds*, the $t = 0$ profiles of solutions of (1) that are polynomial in time. With a constant α these profiles are the polynomials, the initial values of the heat polynomials [RW59]. In one dimension, with $L = \partial_x \alpha \partial_x$ and anchored at a stencil's evaluation point x_e , they form the chain

$$(2) \quad \phi_0 = 1, \quad \alpha \phi_1' = \alpha(x_e), \quad L\phi_k = k(k-1) \alpha(x_e) \phi_{k-2} \quad (k \geq 2),$$

with $\phi_k(x_e) = 0$ for $k \geq 1$ and $\phi_k'(x_e) = 0$ for $k \geq 2$, integrated through the edge as a first-order system of ordinary differential equations that never differentiates α (Section 3.1). The seed ϕ_1 has a constant flux, and ϕ_2 is

the profile of a solution whose rate of change u_t is constant; in general the even seeds carry $\partial_t^k u = \text{const}$ and the odd ones $\alpha \partial_x \partial_t^k u = \text{const}$, which are the interface conditions of a jump. Read across a jump, the chain is the translated polynomial basis of the interface-aware stencils (Section 3.2), so one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension it is classical (Section 1.1). It is also the construction of our companion paper for the wave equation [Mar26], specialised to a single field: the two interface conditions of diffusion, on u and on its flux, are both conditions on u , so one chain serves every stencil.

In two dimensions the seeds replace the polynomial augmentation of the interface-aware RBF-FD stencils, and the Gaussian part and the saddle-point solve stay as they were. Where α varies along the normal only, as across a straight edge, seeds polynomial in the tangential coordinate turn the chain into a triangular system of ordinary differential equations in the normal coordinate, one short march per stencil (Section 5.1). Where the edge is curved or α varies along it, a profile frozen along the normal is inconsistent at $O(1)$ through an unresolved edge, and the chain is marched in the curve's own coordinates instead (Section 5.3). A resistive layer thinner than the stencil, the insulating ring of [MF17b], needs its exact width, its own series and seeds of one degree higher that carry the flux (Section 5.4). An imperfect contact costs strong-form schemes a derivative in the jump conditions, as Lombard and Piroux observed for elastic waves [LP03, LP06]; the flux seeds add the functions of one degree higher that carry the flux, and nothing else.

We measure the seeds in one and two dimensions, on elliptic and parabolic problems, against three baselines: the standard stencils with α sampled at the nodes; the interface-aware stencils of the jump applied at the edge's centre; and the coefficient treatments of diffusion practice, which change the coefficient and keep the scheme: harmonic and arithmetic means of α over a cell (over a disc on scattered nodes), the edge widened to the grid, and, in one dimension, the conservative scheme with exact face conductances. On constant pieces the one-dimensional equilibrium cannot rank these methods, since its solution has a constant flux and lies in the span of the first two seeds, on which the seeds and the exact face conductances are both exact (Section 3.5); the one-dimensional ranking is the parabolic problem's. In two dimensions the tangential variation of the solution takes it out of that span, and the equilibrium is a real test.

Contributions.

- (1) The seeds for the diffusion operator on the unchanged scattered nodes of the interface-aware RBF-FD stencils, which they reduce to as $\delta \rightarrow 0$: the one-dimensional chain (Section 3), its two-dimensional form for a straight edge in the normal coordinate, for a curved or tangentially varying edge in the curve's own coordinates, and for a resistive layer thinner than the stencil (Section 5). To our knowledge

no earlier scheme builds the basis of stencils on an unchanged scattered node set through a smooth material edge thinner than the node spacing by marching ordinary differential equations; its ingredients (the exact face conductance, the formal powers, bases from local solutions of the equation) are each classical, and in one dimension so is the construction.

- (2) In one dimension (Section 4): the knee of the standard operator, a drop of 100 to 200 times between $h = 2\delta$ and $h = \delta/2$ at a 9 : 1 edge and of 770 to 980 times at the 10 : 1 edges of the sinusoidal medium; the jump construction on its floor, the distance of order δ between the jump's solution and the smooth one, while $h \gtrsim 2\delta$ on the two-constant medium, and growing once the grid resolves the edge; and the seeds, fourth order at every δ on one line: on the ramp problem of the two-constant medium they lie within 0.1 to 0.3%, 1 to 1.5% and 3 to 5% of the jump's line at $\delta = 0.0025$, 0.01 and 0.04.
- (3) On scattered nodes through a flat band (Section 6): the seeds are fourth order at every δ tested, elliptic and parabolic, with elliptic fits of 4.23 to 4.48 over 1250 to 40,000 nodes; the elliptic errors at the five widths, the jump included, differ by factors of 1.2 to 2.4 at each node count, and at 40,000 nodes and $\delta = 0.0025$ the seeds' error is 7.99×10^{-9} against the standard operator's 8.36×10^{-5} . That is no deterioration as $\delta \rightarrow 0$ on these node sets, not a proven δ -uniform constant. The rule is to seed every row whose stencil sees an edge within 20δ , with no threshold in δ . The seeded rows' least diagonal dominance never falls below the jump-aware rows', the seeded matrices' condition estimates and GMRES counts stay bounded as $\delta \rightarrow 0$, and their spectra have no eigenvalue in the right half-plane.
- (4) Through the curved band of case 2 the tangential chain is fourth order, within 1.4 to 3.9 times of the flat seeds' RMS error from 5000 nodes at every δ (Section 6.5). On the insulating ring thinned at a fixed resistance from $s = 10^3$ to 10^{11} the seeds are exact on the matched radial profile to rounding (1.3×10^{-15} to 2.1×10^{-15}) and converge at fourth order (fits 4.43 to 4.71) at every s ; through a smooth ring whose width and edges are both below h they converge at fourth order on the nodes away from it (Section 6.6).
- (5) The comparison with the coefficient treatments (Sections 4.5 and 6.7). The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at $\delta = 0$, ahead of the seeds below

801 nodes, where the seeds' line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

- (6) An open implementation in Python, written with the AI tools disclosed at the end of the paper, with a test for every numerical routine, in which every figure and table of this paper is drawn by a committed script from committed results files (<https://github.com/bradleypmartin/rbf-elliptic-parabolic-interfaces-2026>).

1.1. Relation to prior work. In one dimension the construction is classical in every part, and we claim nothing for it. The first seed, the constant-flux function $\alpha_e \int d\xi/\alpha$, is over one cell the exact face conductance of Tikhonov and Samarskii's homogeneous schemes [TS62, Sam01] and, for a piecewise-constant coefficient, the harmonic-mean interface conductivity of finite-volume heat conduction [Pat80]; the conservative scheme built on it, exact at the source-free equilibrium, is second order [PXL26]. The whole chain is the representation of Sturm–Liouville solutions by recursive integrals, as power series in the spectral parameter [Kra08, KP10], taken from the anchor; for a constant coefficient its time-polynomial solutions are the heat polynomials [RW59, Wid75]. Three-point schemes of any order for $(ku)'$ with a piecewise-smooth k , whose coefficients come from Cauchy problems integrated across each cell, are the exact and truncated difference schemes [SM90, GHMK11]; three-point schemes of any order exist across a jump too [Gar89], one of them built by carrying Taylor data across the step [Suj10]. In two dimensions, local solutions of the equation that satisfy the interface conditions are the basis of the Trefftz difference schemes of FLAME, on irregular stencils too [Tsu06, Tsu10], and of the tailored finite point methods [Hua09, HZ12]; Galerkin methods build their bases from local solves [BCO94, HW97, CGH10, OZ07, MP14] and reach high order through coefficients the mesh does not resolve [Mai21, HP23], and the quasi-Trefftz spaces, one of them for the heat equation, build polynomial local solutions by Taylor expansion [IGMPS25, GPS26], which cannot reach through an edge thinner than the stencil [Mar26]. For a jump, the interface-aware RBF-FD stencils this work generalises [Mar16, MF17b] stand beside the immersed interface method and unfitted finite elements [LL94, LI06, HH02, BCH⁺15], now of fourth and sixth order [FZ20, FHM24], all of which take the jump as given. A resistive layer thinner than the grid is an imperfect interface in the limit [SP74, PP13, Has01], which Section 5.4 takes up. The standing alternative changes the coefficient rather than the stencil: harmonic or arithmetic means over a cell [Pat80, EIL01, GA92], averaging for layering finer than the grid [MDH⁺99, DDH03], or a regularised coefficient [TE03, TE06]. Those methods take the cell or the width from the grid; here the width is a property of the medium, below the node spacing, and the aim is the scheme's full order through it. We found no prior instance of stencils

on an unchanged scattered node set whose basis is continued through a sub-grid smooth edge by ordinary differential equations, nor of such stencils reducing to a jump construction as $\delta \rightarrow 0$; the search is logged in the repository's `LITERATURE.md`. We ran the cell means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, as our implementation of each on one problem per dimension: through an edge the grid does not resolve none is better than second order, and on the scattered nodes none improves on sampling the coefficient by more than a factor of 1.75 in the root-mean-square error. We claim nothing beyond that measurement.

Outline. Section 2 fixes the problems, the discretisations and the interface-aware stencils of a jump that the seeds extend. Section 3 constructs the seeds in one dimension and proves what can be proved about them. Section 4 measures them against the standard operator, the jump construction and the coefficient treatments. Section 5 builds the seeds on scattered nodes for straight, curved and doubly sub-grid features. Section 6 measures them there, elliptic and parabolic, with the matrices they make. Section 7 records the limitations and what we would do next.

2. SETTING

Notation. The symbols the sections share, fixed here:

n, h	node count and spacing: $h = 2/(n-1)$ in one dimension and the row spacing $1/\text{round}(0.95\sqrt{n})$ in two
$\alpha; \alpha^-, \alpha^+; L$	conductivity; the pieces on the two sides of an edge; $L = \nabla \cdot (\alpha \nabla \cdot)$
$\delta; \sigma$	width of a tanh edge, $\delta = 0$ the jump; the blend weight $\sigma(z) = \frac{1}{2}(1 + \tanh z)$
s	the thinness of the ring of [MF17b], $1/s$ its width
$x_e; (x', y')$	a stencil's evaluation point (anchor); its local frame, y' normal to the edge
ϕ_k	the one-dimensional seeds (2)
$D_x, D_y; A$	first-derivative matrices; $\text{diag } \alpha$ at the nodes
naive, direct	$D_x A D_x (+D_y A D_y)$; $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$ on one stencil
jump-aware	the interface-aware stencils of Section 2.4
the $\delta = 0$ construction	the same, applied at a smooth edge's centre
seeds	the operator whose rows that see an edge are rebuilt on seeds

2.1. The problem and its interface conditions. On a domain Ω in one or two dimensions we solve (1) with Dirichlet data on $\partial\Omega$, and its equilibrium $Lu = 0$, the elliptic problem. The conductivity $\alpha > 0$ is one smooth function on each side of every interface Γ . Across a jump the temperature and the heat flux are continuous at all times, hence so are all their time derivatives,

and $\partial_t^k u = L^k u$ turns these into conditions on spatial derivatives,

$$(3) \quad [L^k u] = 0, \quad [\alpha \partial_n L^k u] = 0 \quad \text{on } \Gamma, \quad k = 0, 1, 2, \dots,$$

where $[\cdot]$ is the difference of the two sides' limits, n the normal to Γ , and each side's L is taken with that side's α [Mar16, MF17b].

A smooth edge. With an edge width $\delta > 0$ the jump at an interface is replaced by a tanh transition between the two pieces,

$$(4) \quad \alpha = (1 - \sigma(d/\delta)) \alpha^- + \sigma(d/\delta) \alpha^+, \quad \sigma(z) = \frac{1}{2}(1 + \tanh z),$$

where d is the signed distance to the interface, $x - x_c$ in one dimension and the distance along the normal through the foot point on the curve in two, positive on the side of α^+ . The pieces keep their own variation, and $\delta = 0$ is the jump. A band between two interfaces is blended twice, from the outside piece into the inside one across the lower interface and back across the upper, so that a band wider than its edges has two independent transitions; Section 5.4 defines the smooth form of a layer thinner than its edges. Beyond 19δ from its centre an edge is its pieces to within two units in the last place.

Three regimes. For $\delta \gg h$ a stencil sees a smooth coefficient. For $\delta = 0$ it sees a jump, and (3) applies. For $0 < \delta \lesssim h$ it sees what looks to it like a jump between two of its nodes, but no interface condition applies, since α is smooth, and the smooth problem and the jump problem have different solutions. They differ at first order in δ : in one dimension the resistance $\int dx/\alpha$ of a tanh edge between constant pieces a and b falls short of the jump's by

$$(5) \quad c\delta, \quad c = \frac{(a-b)\ln(a/b)}{2ab},$$

which is 8.79δ for the 9 : 1 edge of Section 2.2 and is positive for either ordering of a and b .

Norms and orders. In one dimension an error is the relative ℓ^2 error at the nodes, $\|e\|_2/\|u\|_2$, and a rate is $\log_2$ of the ratio of consecutive errors as the node count doubles. In two dimensions an error is the root-mean-square (RMS) error over all nodes, the Dirichlet nodes included, and an order is the least-squares slope of $\log e$ against $\log h$ over the node counts named. A maximum-norm error is named as such wherever one is quoted, and every caption says which norm it shows.

2.2. One dimension. The one-dimensional problem is (1) with $L = \partial_x \alpha \partial_x$ on $[-1, 1]$ with Dirichlet ends. A grid has n equispaced nodes with both ends on nodes, so $h = 2/(n-1)$. Two media are used, each at $\delta = 0, 0.04, 0.01$ and 0.0025 :

- the *two-constant medium*, $\alpha = 1/9$ for $x < 0$ and 1 for $x > 0$, the single interface of the time-dependent test in the code behind [Mar16], which the dissertation does not report; the edge lies midway between two nodes when n is even;

- the *sinusoidal medium* of [Mar16, eq. (75)], $\alpha = 0.1 + 0.4 \sin 2\pi x$ on the layer $[0, \frac{1}{2}]$ and 1 elsewhere, whose two edges both have the contrast 10 : 1, since the sinusoid vanishes at both; both edges fall on nodes when $n - 1$ is a multiple of 4.

Each medium carries two problems: the equilibrium with $u(-1) = 1$ and $u(1) = 0$, and the *ramp problem*, which starts from $u = 0$, holds $u(1, t) = 0$, raises $u(-1, t)$ from 0 to 1 over $0 \leq t \leq 1$ by the C^∞ step $e^{-1/t}/(e^{-1/t} + e^{-1/(1-t)})$, and is read at $t = 2$. (The original code ramps with a logistic function, whose value and derivatives at $t = 0$ are small but not zero; the mismatch with the zero start at the corner would cap a fourth-order study.) Section 4.1 gives the reference solutions.

The *naive* operator is $D_x A D_x$ [Mar16, eq. (76)], with D_x the fourth-order first-derivative matrix of Fornberg’s weights [For88], one-sided at the ends, and $A = \text{diag } \alpha(x_j)$, which gives nine-point rows in the interior. It is the form in which a standard stencil sees a jump at all: a direct discretisation of $\alpha u_{xx} + \alpha' u_x$ with the coefficients sampled at the nodes is blind to the jumps, and its error on the sinusoidal medium stays at 0.22 as the grid is refined. Time stepping is the fourth-order backward differentiation formula (BD4) with $\Delta t = h$, one sparse LU factorisation per operator and the Dirichlet values imposed at every step; its three starting values come from classical Runge–Kutta sub-steps at half the stability limit.

2.3. Two dimensions. The two-dimensional problems are the three test cases of [MF17b], (1) on the unit strip $0 \leq y \leq 1$, periodic in x with period 1, with Dirichlet data on $y = 0$ and $y = 1$. The conductivity is one smooth piece in a band between two interfaces and another outside it.

- *Case 1*, a flat band $0.6 \leq y \leq 0.8$ with $\alpha = 0.2$ in it and 1 outside. With $u = \sin 2\pi x$ on $y = 1$ and $u = 0$ on $y = 0$ the equilibrium is $\sin 2\pi x v(y)$ with v piecewise exponential, and the parabolic problem is marched to $t = 0.1$ against the exact solution $e^t \sin 2\pi x w(y)$ of the same form [MF17b].
- *Case 2*, both interfaces bent to $y = 0.6 + 0.02 \sin 2\pi x$ and $y = 0.8 + 0.02 \sin 2\pi x$,¹ with $\alpha = 0.2 + 0.1 \sin 2\pi x \sin 2\pi y$ in the band and the boundary data of case 1.
- *Case 3*, a ring $0.35 - 1/s \leq r \leq 0.35$ about $(\frac{1}{2}, \frac{1}{2})$ with $\alpha = 1/(1.5s) + \sin 2\pi x \sin 2\pi y/(3s)$ in it and 1 outside, around a cooling circle $r = 0.05$ whose inside is removed, with $u = \sin 6\pi x$ on $y = 0$ and $y = 1$ and $u = 0$ on the circle. At $s = 10^3$ this is the insulating ring of [MF17b], 0.001 wide at a contrast of about 1500 : 1; the extremizing sweep of that paper runs s up to 10^{11} , thinning the ring at a fixed resistance, its width over its conductivity, of $1.5/(1 + \frac{1}{2} \sin 2\pi x \sin 2\pi y)$.

¹The published texts do not state the curves; these are the ones of the original code.

With $\delta > 0$ each interface of cases 1 and 2 becomes an edge (4) in the signed normal distance to its curve; the ring's smooth form is Section 5.4's. On case 1 the medium is then the one-dimensional one in y , and the reference solutions stay separable (Section 6.1).

Node sets. A set of n nodes is built as in [MF17b]. Rows hold $m = \text{round}(0.95\sqrt{n})$ nodes per unit length, so their spacing is $h = 1/m$. Six fixed rows straddle each interface, three on each side at normal distances $(\frac{1}{2} + j\frac{\sqrt{3}}{2})h$, $j = 0, 1, 2$, the middle row on each side staggered by half a spacing along the curve, so that the rows cross the interface in a hexagonal pattern; a fixed row lies on each Dirichlet curve; and the remaining nodes are placed by 100 steps of a simulated repulsion from a random start, kept clear of the straddling rows. The ring is thinner than h at every node count used, so one set of six rows straddles its midline, and both of its interfaces pass between the innermost pair. Unless a spread over node sets is quoted, each node count has one node set, from a fixed random seed.

RBF-FD. RBF-FD [FF15a, FF15b] builds each row of a differentiation matrix from a stencil. For a stencil of nodes $x_1, \dots, x_{N_s}$ about an evaluation node x_e , the weights w of a linear operator $\mathcal{L}$ solve

$$(6) \quad \begin{bmatrix} \Phi & P \\ P^\top & 0 \end{bmatrix} \begin{bmatrix} w \\ w_* \end{bmatrix} = \begin{bmatrix} \mathcal{L} \varphi(\|x - x_i\|)|_{x_e} \\ \mathcal{L} p_j(x)|_{x_e} \end{bmatrix},$$

with $\Phi_{il} = \varphi(\|x_i - x_l\|)$ for the Gaussian $\varphi(r) = e^{-(\varepsilon r)^2}$, $P_{ij} = p_j(x_i)$ over the bivariate monomials p_j up to a degree p in coordinates scaled by the stencil's radius, and w_* discarded. The constraint rows make the weights exact on the polynomials, which take over from the Gaussians under refinement [FFBB16, BFFB17]; the shape parameter is $\varepsilon = 0.4/d$, with d the distance from x_e to its nearest neighbour. Stencils have 42 nodes and degree 5 in the bulk, and 30 nodes and degree 4 within $3/\sqrt{n}$ of a Dirichlet curve and wherever a 42-node stencil would reach across an interface. Of the latter, the interface-aware construction rebuilds the rows whose 30 nodes cross an interface (Section 2.4); the seeds rebuild the rows whose stencils see an edge (Section 5.2).

Operators and solvers. The naive operator is $D_x A D_x + D_y A D_y$, the two-dimensional form of [Mar16, eq. (76)], with every node's rows of D_x and D_y built on its own stencil, so that it reaches the neighbours of the neighbours. The direct operator applies $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$ at each node through one stencil; it is blind to a jump and serves as the control where the grid resolves the edge. Dirichlet rows are the identity. The elliptic systems are solved directly with SuperLU; the iterative study applies GMRES and BiCGSTAB to the interior system, with and without an incomplete LU factorisation and the diagonal-dominance preconditioner of [Mar16, App. B]. The parabolic problem of case 1 is marched by BD4 with $\Delta t = h$ from the exact solution's history, 3 to 19 steps to $t = 0.1$.

2.4. Interface-aware stencils for a jump. The seeds extend the interface-aware stencils of [Mar16, MF17b], whose wave-equation form is in [MFSC15, MF17a]. We summarise them in the form the later sections use; the derivations are in those references.

One dimension. Expand u on each side of an interface x_c in a polynomial of degree four about x_c , with the Taylor coefficients of that side's α entering through multiplication matrices on the coefficient vectors, and impose (3) on $L^k u$ for $k = 0, 1, 2$ and on the flux for $k = 0, 1$. These five conditions give a square *continuity matrix* per side, $C^- a^- = C^+ a^+$ on the coefficient vectors $a^\pm$. Keeping the monomials on the evaluation node's side and carrying them across with $(C^+)^{-1} C^-$ gives the *translated basis*: piecewise polynomials, one per monomial, that satisfy the five conditions. The weights of a five-node stencil that straddles x_c reproduce L at the evaluation node on this basis, a 5×5 solve per stencil, and a stencil that straddles both edges of a thin layer translates twice.

Two dimensions. A stencil whose 30 nodes lie in more than one region is given a local frame at the point of each interface nearest its centre, x' along the tangent and y' along the normal, in units of the stencil's radius, and the interface is expanded there as $y' = f(x')$. The continuity conditions equate, along the interface, the first $5 - 2k$ coefficients in x' of $L^k u$ and the first $4 - 2k$ of the normal flux $\alpha(\partial_{y'} - f' \partial_{x'}) L^k u$, fifteen conditions for degree four, one per bivariate monomial; the curvature enters through f , in the curve along which the coefficients are read and in the normal derivative. The translated basis replaces the monomials p_j in (6). The Gaussians are *warped*: in the frame of the evaluation node's side, the normal coordinate of every node is replaced by a piecewise-linear one whose slope changes by the factor α^-/α^+ across each interface, so that a Gaussian's value and its $\alpha \partial_n$ are both continuous there and the RBF part of the stencil satisfies the flux condition to first order.

This construction, with warped Gaussians, is what we call *jump-aware*. Applied with the centre of a smooth edge as the interface and the pieces' Taylor data as if δ were 0, it is the $\delta = 0$ *construction* of Sections 4.3 and 6.2. It is also the $\delta \rightarrow 0$ limit of the seeds, which reproduce it at $\delta = 0$ (Sections 3.2 and 5.2).

2.5. Scope of the implementation. The experiments run on a reimplementation of the 2016–2017 methods in Python (NumPy and SciPy, with a test for every numerical routine), written for this study. Every published figure of the one- and two-dimensional heat problems of [Mar16, MF17b] but the cornered interface's has a counterpart in it, and every published convergence line is reproduced at its published rate or better. Where the numbers part company the repository's port notes say so. Two differences bear on this paper: the iterative solvers do not break down on the ring here (BiCGSTAB converges at every node count), and the extremizing sweep does not lose convergence at $s = 10^{11}$ (every s is fourth order down to its

reference's own error). The 2016 two-dimensional code is not preserved, so neither difference could be traced to it.

The implementation leaves out the cornered interface of [Mar16, Sec. 5.4.5], which is approximated there by circular segments at second order. Its two-dimensional media are a band between two curves, periodic in x , so no edge meets the domain's boundary. The bulk stencils are degree five and the stencils that see an edge degree four, seeded or interface-aware. Each case has one contrast: 9 : 1 and 10 : 1 in one dimension, 5 : 1 on cases 1 and 2, and on the ring about 1500 : 1 at $s = 10^3$, growing with s at a fixed resistance. Every smooth edge is a tanh. Section 7 returns to these restrictions.

3. SEEDS FOR THE DIFFUSION OPERATOR IN ONE DIMENSION

Fix a stencil of the grid of Section 2.2: five nodes $x_1 < \dots < x_5$ and an evaluation point x_e , one of them. The standard weights are exact on the monomials $(x - x_e)^k$, $k \leq 4$, and the interface-aware weights of Section 2.4 on the translated basis. This section builds the basis for a smooth edge, shows that the translated basis is its limit as $\delta \rightarrow 0$, proves that every stencil solve is nonsingular, and says what the truncation error is. Each ingredient is classical in one dimension (Section 1.1), and the construction is that of our companion paper for the wave equation [Mar26], specialised to the diffusion operator.

3.1. Seeds as the profiles of solutions polynomial in time. Look for solutions of $u_t = Lu$, $L = \partial_x \alpha \partial_x$, that are polynomial in time, $u = \sum_{j=0}^J t^j g_j(x)$. Matching powers of t gives $(j+1)g_{j+1} = Lg_j$ and $Lg_J = 0$, so the profile at $t = 0$ satisfies $L^{J+1}g_0 = 0$. The heat equation spends one power of L on each power of t where the wave equation spends two, but the profiles fill the same nested kernels of L . With a constant α they are the polynomials: the time-polynomial solutions of $u_t = \alpha u_{xx}$ are the heat polynomials

$$v_k(x, t) = \sum_{0 \leq j \leq k/2} \frac{k!}{j!(k-2j)!} x^{k-2j} (\alpha t)^j$$

[RW59, Wid75], whose profiles at $t = 0$ are the monomials. The seeds are the same profiles for a variable α , anchored at the evaluation point.

Definition 3.1 (Seeds). Let $\alpha > 0$ and $\alpha_e = \alpha(x_e)$. The *seeds* at x_e are the functions $\phi_0, \phi_1, \dots$ of the chain (2): $\phi_0 = 1$, $\alpha \phi'_1 = \alpha_e$ and $L\phi_k = k(k-1)\alpha_e \phi_{k-2}$ for $k \geq 2$, with $\phi_k(x_e) = 0$ for $k \geq 1$ and $\phi'_k(x_e) = 0$ for $k \geq 2$.

Lemma 3.2. *For every k the function*

$$(7) \quad u_k(x, t) = \sum_{0 \leq j \leq k/2} \frac{k!}{j!(k-2j)!} (\alpha_e t)^j \phi_{k-2j}(x)$$

solves $u_t = Lu$, and $u_k(\cdot, 0) = \phi_k$. With a constant α , $\phi_k = (x - x_e)^k$ and $u_k(x, t) = v_k(x - x_e, t)$.

Proof. Write $c_j = k!/(j!(k-2j)!)$. By the chain, $Lu_k = \sum_j c_j (k-2j)(k-2j-1) \alpha_e^{j+1} t^j \phi_{k-2j-2}$ and $\partial_t u_k = \sum_j (j+1) c_{j+1} \alpha_e^{j+1} t^j \phi_{k-2j-2}$, and the two agree because $(j+1) c_{j+1} = c_j (k-2j)(k-2j-1)$. With α constant, $L(x - x_e)^k = \alpha k(k-1)(x - x_e)^{k-2}$ and the anchor conditions fix the two constants of integration, so $\phi_k = (x - x_e)^k$ by induction; with $\alpha_e = \alpha$, (7) is then $v_k(x - x_e, t)$. $\square$

Read one at a time, the seeds say what each monomial becomes through the edge. The seed ϕ_1 solves $L\phi_1 = 0$: it is the equilibrium of constant flux, $\phi_1(x) = \alpha_e \int_{x_e}^x d\xi/\alpha(\xi)$, and what stays smooth through the edge is its flux $\alpha\phi_1'$, not its slope. The seed ϕ_2 is the profile of $u_2 = \phi_2 + 2\alpha_e t$, whose rate of change is constant, $u_t = 2\alpha_e$; the constant matters, since $u_t = 0$ would only return $\ker L$. In general, by (7), $\partial_t^m u_{2m} = (2m)! \alpha_e^m$ is constant and $\partial_t^m u_{2m+1} = (2m+1)! \alpha_e^m \phi_1$ has a constant flux: the even seeds carry $\partial_t^m u = \text{const}$ and the odd ones $\alpha \partial_x \partial_t^m u = \text{const}$. Taken in the order $\phi_0, \dots, \phi_4$ these are the five conditions of (3) from which the translated basis of a jump is built, on u , its flux, $u_t = Lu$, the flux of u_t , and $u_{tt} = L^2 u$. In the wave equation the same alternation passes between two fields; in diffusion both kinds of condition bear on u , so one chain serves every stencil.

In one dimension the seeds are a classical object. Integrating the chain twice from the anchor gives $\phi_k(x) = k(k-1) \alpha_e \int_{x_e}^x \alpha(s)^{-1} \int_{x_e}^s \phi_{k-2}(r) dr ds$, and unrolling the recursion,

$$(8) \quad \phi_k = k! \alpha_e^{\lceil k/2 \rceil} I_k, \quad I_k(x) = \int_{x_e}^x \frac{ds_1}{\alpha(s_1)} \int_{x_e}^{s_1} ds_2 \int_{x_e}^{s_2} \frac{ds_3}{\alpha(s_3)} \dots,$$

where $I_0 = 1$ and I_k has k nested integrals whose weights alternate between $1/\alpha$ and 1, starting from $1/\alpha$ outermost. These are the recursive integrals, $X^{(k)}$ for odd k and $\tilde{X}^{(k)}$ for even k , by which Kravchenko and Porter write the solutions of $(pu')' + qu = \lambda ru$ as power series in the spectral parameter [Kra08, KP10], here with $q = 0$, $r = 1$, $p = \alpha$, the particular solution $u_0 = 1$ and the integrals taken from the anchor. The seeds are thus the formal powers of the Sturm–Liouville operator L anchored at the evaluation point, and the coefficients of those series are the time-polynomial profiles, with λ in the role of ∂_t . The repository’s tests check the identity against the ODE march below to a relative 10^{-9} through a tanh edge.

The weights w_i of a stencil approximate Lu at x_e from the nodal values by reproducing L on the first five seeds,

$$(9) \quad \sum_{i=1}^5 w_i \phi_k(x_i) = (L\phi_k)(x_e) = \begin{cases} 2\alpha_e, & k = 2, \\ 0, & k = 0, 1, 3, 4, \end{cases}$$

since $L\phi_k = k(k-1) \alpha_e \phi_{k-2}$ vanishes at x_e for every k but 2. These are the conditions the interface-aware stencils impose on the translated basis, and

with a constant α they return α times the standard second-derivative weights [For88], which the implementation reproduces to rounding. Stencils exact on a local space other than the polynomials go back to fitted operators and to HODIE [AS55, Il'69, LR80, RST08]. Three-point schemes whose coefficients come from Cauchy problems integrated across a piecewise-smooth coefficient are the exact and truncated schemes [SM90, GHMK11], and the seed stencils are a five-point variant of them.

First-order form. The chain is integrated as a first-order system in each seed and its flux $\psi_k = \alpha \phi'_k$,

$$(10) \quad \phi'_k = \psi_k / \alpha, \quad \psi'_k = k(k-1) \alpha_e \phi_{k-2},$$

with $\phi_0(x_e) = 1$, $\psi_1(x_e) = \alpha_e$ and every other initial value zero, so that α is never differentiated. The ten states march together from x_e outward to each side of the stencil. Across a jump ϕ and ψ are continuous, which is L read in the weak sense: the march stops at the jump and reads α on each side from the piece that owns it, so a jump is marched like any other width, and a stencil that straddles both edges of a thin layer marches through both.

Stencil coordinate. In the coordinate $\xi = (x - x_e)/h_s$, with $h_s = \max_i |x_i - x_e|$, the functions $\phi_k(x)/h_s^k$ satisfy the same chain with $\alpha(x_e + h_s \xi)$ in place of α . They are close to ξ^k , and the matrix $A_{ki} = \phi_k(x_i)/h_s^k$ of (9) is a perturbed Vandermonde matrix; the weights are $w = \tilde{w}/h_s^2$, with $A\tilde{w}$ equal to the right-hand side of (9). For the centred stencil, $\xi_i = 0, \pm\frac{1}{2}, \pm 1$, the condition number of A is 23.5 with a constant α . On the stencil of Figure 1, half a cell from a 9 : 1 edge, it is 90 at $\delta = 0$ and 138 at $\delta = h/2$, and 24 again once $\delta \geq 10h$, where the seeds are close to the monomials.

The march and the seeded rows. The march is an explicit adaptive Runge–Kutta method of order eight (SciPy’s DOP853) at relative tolerance 10^{-13} and absolute tolerance 10^{-15} , with segment boundaries at each edge centre and at 10δ on either side of it, so that the step never has to find the edge inside a long step over flat material. A row is seeded when the span of its stencil meets $(x_c - 20\delta, x_c + 20\delta)$ for an edge centre x_c , beyond which the edge is its pieces to the last bit; every other row is the standard one. At $\delta = 0$ the seeded rows are exactly those that straddle the jump; at $\delta = 0.0025$ and 0.04 on 100 nodes they number 8 and 84; once 20δ covers the domain, every row is seeded. No threshold in δ switches the seeds on or off: a resolved edge costs them nothing (Section 4.4), and at $\delta = 0$ they are the jump’s weights (Section 3.2). Rows whose nodes sit at the same offsets from x_e march as one system, so an operator costs some sixteen calls of the integrator whatever the number of seeded rows: 0.11 s for the 1284 seeded rows of $\delta = 0.04$ on 1600 nodes. The march has one floor of its own. With L_h the discrete operator, the seeded rows’ residual on the exact equilibrium, $h^2 \max |L_h u|$, is at most 2.5×10^{-16} at 200 nodes while the march crosses the edge in a few steps ($\delta \geq h/10$), and 2.4×10^{-13} and 5.8×10^{-13} at $\delta = h/100$ and $h/1000$, where it takes many. The equilibrium solution shows it once

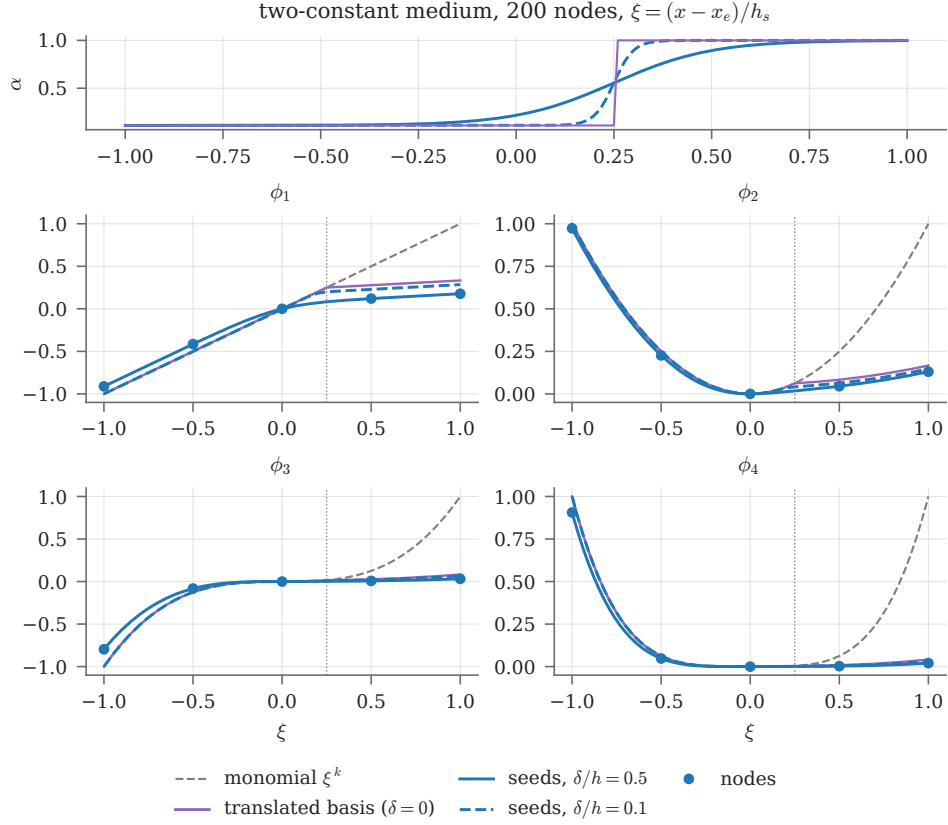


FIGURE 1. The seeds $\phi_1, \dots, \phi_4$ of one five-point stencil of the two-constant medium at 200 nodes, whose evaluation point x_e lies half a cell to the left of the 9 : 1 edge (dotted), in the stencil coordinate $\xi = (x - x_e)/h_s$ with $h_s = 2h$; α across the stencil on top. Grey, dashed: the monomials ξ^k . Purple: the translated basis of the jump ($\delta = 0$). Blue: the seeds at $\delta = h/2$ (solid, with their values at the nodes) and $\delta = h/10$ (dashed). Each seed is multiplied by $(\alpha_0/\alpha_e)^{\lceil k/2 \rceil}$, with α_0 the jump's value at x_e and α_e the edge's, so that the lines compare across δ ; the factor is the normalisation of (2), which the weights never see. On the node's side of the jump its seeds are the monomials, and across it the slope of ϕ_1 falls ninefold; the seeds of a smooth edge bend through it instead of kinking, and approach the translated basis at first order in δ/h .

$\delta \lesssim h/40$, at 10^{-11} to 4×10^{-11} . A collocation integration of the same chain would remove the floor; we have not built one.

3.2. Nested kernels and the jump as the limit. Since L sends each seed two steps down the chain, $\text{span}\{\phi_0, \dots, \phi_k\}$ is the $(k+1)$ -th space of

$$(11) \quad \{1\} \subset \ker L \subset \ker \partial_x L \subset \ker L^2 \subset \ker \partial_x L^2 \subset \dots,$$

where $\ker \partial_x L^m$ holds the g with $L^m g$ constant. The spaces depend on neither the anchor nor the normalisation, so the weights do not either; only the basis does.

For a jump, read L in the weak sense, ϕ and $\alpha\phi'$ continuous across it. Then $\ker L$ is spanned by 1 and the piecewise-linear function whose slope changes by the factor α^-/α^+ across the jump, the translated x of Section 2.4. On constant pieces $L\phi_2 = 2\alpha_e$ makes $\phi_2'' = 2\alpha_e/\alpha^\pm$ on the two sides, with ϕ_2 and its flux continuous, which is the continuity of u_t and of its flux; and so on up the chain, each condition $\partial_t^m u = \text{const}$ or $\alpha \partial_x \partial_t^m u = \text{const}$ being one of (3). For constant pieces the chain across a jump is therefore the translated basis, the anchor conditions picking one basis of its span, and the interface-aware stencils are the seed stencils at $\delta = 0$: one construction with two implementations, algebra for a jump and an ODE march for a smooth edge. On the stencil of Figure 1 the march reproduces the translated basis to 3×10^{-15} function by function, and its weights to 7×10^{-16} . The same device, Taylor data carried across a step to make the weights, appears in photonics [Suj10], and three-point schemes of any prescribed order across a jump in the coefficient are Gartland's [Gar89].

As $\delta \rightarrow 0$ the seed weights approach the jump's at first order in δ/h . On the stencil of Figure 1, $\max_i |w_i - w_i^{\text{jump}}| / \max_i |w_i^{\text{jump}}|$ is 84, 46, 11, 1.1 and 0.11 percent at $\delta/h = 1, \frac{1}{2}, 0.1, 0.01$ and 0.001, and the seeds approach the translated basis at the same rate. A stencil that straddles both edges of a layer two cells thick, $\alpha = 1 | 0.1 | 1$, marches through both: at $\delta = 0$ its weights are those of the jump construction translated twice, to 7.6×10^{-15} , and at $\delta/h = 0.1, 0.01$ and 0.001 they differ from them by 22, 1.9 and 0.18 percent.

When the pieces vary the limit is not exact. The interface-aware construction expands each side's α to degree four about the interface and truncates the operator there (Section 2.4), while the seed ϕ_2 solves $(\alpha\phi_2')' = 2\alpha_e$ exactly, $\phi_2 = (x - x_e)^2 - \frac{2}{3} (\alpha'(x_e)/\alpha_e) (x - x_e)^3 + \dots$, so that within a stencil the two five-dimensional spaces differ at relative order $h\alpha'/\alpha$. On the sinusoidal medium, whose layer has $\alpha'/\alpha = 25$ at its edge, the seed and jump weights of the row whose evaluation point is that edge differ at $h = 0.01$ by 1.25 times the largest weight at $\delta = 0$ and by 1.26 to 2.24 times for δ/h from 0.001 to 1, and the difference falls with h . Neither space is wrong, each being exact on its own five functions; on this medium the seed operator's error is the smaller, by 6 to 44 times at equilibrium and 6 to 38 times on the ramp problem at $\delta = 0$ over 101 to 1601 nodes. There the seeds are measured against the reference solution, never against the jump construction.

3.3. Well-posedness. The stencil solve is nonsingular for every edge width, the jump included, and for every set of distinct nodes, not only the ones we used.

Proposition 3.3. *Let α be positive and bounded away from zero on an interval I containing x_e , and continuous on I except at finitely many jumps, across which L is read in the weak sense. Then for every k the seeds $\phi_0, \dots, \phi_k$ form a complete Chebyshev system on I : interpolation in their span at any $k + 1$ distinct points of I is uniquely solvable. Where α is smooth, as through a tanh edge between smooth pieces, they form an extended complete Chebyshev system. In particular (9) has a unique solution for every $\delta \geq 0$ and every five distinct nodes in I .*

Proof. By (8) the seeds are nonzero multiples of $I_0, \dots, I_k$, nested integrals whose weights $v_1 = 1/\alpha$, $v_2 = 1$, $v_3 = 1/\alpha, \dots$ (outermost first) are positive. Such a family is the canonical basis of the kernel of the operator in Pólya form $\partial_x \frac{1}{v_k} \partial_x \dots \frac{1}{v_2} \partial_x \frac{1}{v_1} \partial_x$, which is $\partial_x L^m$ for $k = 2m$ and L^{m+1} for $k = 2m + 1$, and whose kernel is the $(k + 1)$ -th space of (11). Suppose $g = \sum_{j \leq k} c_j I_j$ vanishes at $k + 1$ distinct points of I . Then $g' = v_1 g_1$, where $g_1 = \sum_{1 \leq j \leq k} c_j I'_{j-1}$ is a combination of the k nested integrals $I'_0, \dots, I'_{k-1}$ with the weights $v_2, v_3, \dots$, and g_1 is continuous. Between two consecutive zeros $a < b$ of g , $\int_a^b v_1 g_1 = 0$ with $v_1 > 0$, so g_1 vanishes in (a, b) : it has k distinct zeros. By induction on k , $g_1 \equiv 0$, so $c_1 = \dots = c_k = 0$, and then $g = c_0$ vanishes, so $c_0 = 0$. Only the positivity and integrability of the weights entered, so the argument holds across finitely many jumps. Where α is smooth the operator is disconjugate on I and the family is an extended complete Chebyshev system [P622, KS66, Cop71]. $\square$

When x_e lies on a jump, as it does for the rows whose evaluation node falls on an edge of the sinusoidal medium, α_e is the value of the piece that owns x_e ; ϕ_1 then has slope 1 on that side and α_e/α on the other, and nothing else changes, since α_e enters the seeds only as a factor. The argument is the one the companion paper makes for the two fields of the wave equation [Mar26], shortened by (8).

3.4. Truncation error. Interpolation in an extended complete Chebyshev system has a Newton form whose remainder carries the operator that annihilates the space where the polynomial remainder carries the $(k + 1)$ -th derivative [Mü73]. For the five seeds that operator is $M = \partial_x L^2$, and the local truncation error of a seed stencil on a smooth function g is

$$(12) \quad \tau(g) = \sum_{i=1}^5 w_i g(x_i) - (Lg)(x_e) = O(h^3 M g),$$

with a constant that the remainder expresses through $1/\alpha$ inside the stencil. That is third order, not fourth: a second derivative on five points is generically third order, and the centred standard stencil gains its fourth

only because its h^3 term vanishes by symmetry, which an edge breaks. The interface-aware rows of a jump show the same. On the function of degree five in their translated basis, which lies outside the space they reproduce, the h^3 term is 0.32, 0.084 and 172 (in the stencil coordinate) for a 9 : 1 jump half a cell and one and a half cells from the evaluation node and at the node itself, and zero to rounding for equal pieces.

The global order is still four. A local error of order h^3 on a fixed number of rows costs one order less in the solution than the same error on every row, since the inverse of the discrete operator has entries of order h , so the solution error is $O(h^4)$. This is what the interface-aware stencils reach at a jump (rates 3.85 to 4.19, elliptic and parabolic, in our port of [Mar16]), and it is the phenomenon Beale and Layton prove for interface corrections, where an $O(h)$ truncation error at the interface leaves the solution second order [BL06]. The bound (12) has two readings, and Section 4 is built on them.

Equilibrium. If $Lu = 0$ then $Mu = 0$, and more than that: u has a constant flux, so on every seeded stencil it lies in $\text{span}\{\phi_0, \phi_1\}$, and (9) reproduces Lu there exactly, at any δ and any h (Section 3.5).

Time dependence. If $u_t = Lu$ then $L^2u = u_{tt}$ and $Mu = \partial_x u_{tt}$, the x -derivative of another solution of the heat equation. Since u_{tt} has a continuous flux, $\partial_x u_{tt}$ is bounded independently of δ and changes across the edge by the ratio of the pieces, whatever the width. The standard stencil's error is driven by $u^{(5)}$ instead, which grows like δ^{-4} inside the edge, and the rows of $D_x A D_x$ that see an unresolved edge miss a fixed fraction of the jump in slope, an error of order $1/h$ in Lu . The bound does not make the constant in (12) independent of δ/h , since the remainder involves $1/\alpha$ across the stencil. Measured on a separable solution $e^{ct}v(x)$ of the two-constant medium from 100 to 400 nodes, the seeded rows' residual is third order (rates 3.06 to 3.78), its constants at $\delta = 0.01$ and 0.0025 within a factor of two of each other, and the standard rows' residual is fourth order (3.94 to 4.01).

Diffusion smooths, so for $t > 0$ every $\partial_t^k u$ is itself a solution and the second reading holds for the whole chain. And since $\partial_x(\alpha u_x) = u_t$, the flux changes across an edge of width δ only by $\int u_t dx = O(\delta)$: in diffusion a sub-grid edge is close to a jump at all times. That is why the $\delta = 0$ construction is a serious baseline here (Section 4.3), and why the seeds' margin over it is set by δ .

3.5. The elliptic problem in one dimension. The one-dimensional equilibrium has a constant flux $\alpha u' = B$, so $u = u(-1) + B \int_{-1}^x d\xi/\alpha$, and two constructions are exact on it at every h and every δ when the pieces are constant.

The first is the seed operator. Its seeded rows are exact by the first reading of Section 3.4, and its standard rows see a linear function on a constant piece. At $\delta = 0$ it is the interface-aware operator of the jump.

The second is the conservative three-point scheme with the face conductances

$$(13) \quad a_{i+1/2} = h / \int_{x_i}^{x_{i+1}} \frac{d\xi}{\alpha(\xi)},$$

whose discrete flux $a_{i+1/2}(u_{i+1} - u_i)/h$ equals B at the exact nodal values whatever α is, so that the exact solution satisfies every discrete equation. It is the homogeneous scheme of Tikhonov and Samarskii [TS62, Sam01], whose face coefficient is the conductance of the constant-flux seed over one cell. For a piecewise-constant α it is the harmonic-mean interface conductivity of finite-volume heat conduction, which Patankar derives from the same steady one-dimensional flux [Pat80, §4.2-3] and which is the exact interblock transmissivity of groundwater models for steady one-dimensional flow through blocks of uniform transmissivity [GA92]. Pan, Xu and Li call the scheme the harmonic average method and prove it second order [PXL26]. The exactness belongs to the conservative form: the same harmonic mean sampled at the nodes and used as A in $D_x A D_x$, the like-for-like changed medium of Section 4.5, leaves a residual above 0.5 on the exact equilibrium of the 9 : 1 jump at 100 nodes, whether the mean is taken over one cell or two.

So on constant pieces the one-dimensional equilibrium cannot rank the seeds against the conservative scheme. It is the sanity check and the limit case, and the one-dimensional ranking is the parabolic problem's, where the conservative scheme is second order and the seeds are fourth (Section 4.5). Both exactness results are classical, and so is the conservative scheme's second order; we observe them and claim none of them. Where the pieces vary, as in the sinusoidal medium, the equilibrium does rank the two, for a reason that has nothing to do with the edge: the conservative scheme stays exact, while the seed operator's standard rows inside the layer keep the error of a smooth coefficient, and it is exact only where the seeded rows cover the layer (Section 4.4).

4. ONE-DIMENSIONAL RESULTS

This section measures the naive operator, the $\delta = 0$ construction and the seeds, and then the coefficient treatments, on the two media of Section 2.2 at $\delta = 0, 0.04, 0.01$ and 0.0025 . Node counts double from 50 to 1600 on the two-constant medium and from 101 to 1601 on the sinusoidal one, whose edges then fall on nodes; $h = \delta$ falls at $n = 51, 201$ and 801 for the three widths. Errors are relative ℓ^2 errors at the nodes and rates are per doubling of n (Section 2.1). The ramp problem is marched by BD4 with $\Delta t = h$, so its errors are those of the fully discrete scheme; Section 4.4 separates the time integrator's share.

4.1. Media and references. With $\delta > 0$ each medium blends its pieces across each edge by (4), the two edges of the sinusoidal medium in turn, so

that the pieces' weights remain a partition of unity; at $\delta = 0$ the blend is the jump to the last bit. Both references cut $[-1, 1]$ into elements on which α is smooth up to their ends: the pieces for a jump, and for an edge also the points $x_c \pm m\delta$, $m = 1, 3, 9, 27$, so that outside the core $|x - x_c| \leq \delta$ each element spans at most a factor of three in distance from the centre and is resolved by a polynomial of modest degree.

The equilibrium reference is $u = u(-1) + B \int_{-1}^x d\xi/\alpha$, with B fixed by $u(1)$ and the integral taken by Gauss quadrature on the elements. Through a tanh edge between constants a and b the integrand has the antiderivative

$$\delta \left[\frac{z}{a} + \frac{a-b}{2ab} \ln(a + b e^{2z}) \right], \quad z = \frac{x - x_c}{\delta},$$

which the quadrature reproduces to 10^{-14} ; subtracting the jump's z/a and z/b on the two sides leaves the deficit (5) as $z \rightarrow \pm\infty$. The parabolic reference is Chebyshev collocation on the same elements, 20 points on each and elements wider than 0.1 split, with u and αu_x matched at every element end, integrated to $t = 2$ by SciPy's implicit Radau method at relative and absolute tolerances 10^{-9} and 10^{-11} ; the absolute tolerance sits above the round-off of the collocation system on elements of width of order δ , which a tighter tolerance would make the step control chase. Against a second resolution, 24 points and elements to 0.05, it agrees to 5×10^{-12} on the two-constant medium and 6×10^{-11} on the sinusoidal one, at worst over the four widths. That is round-off, not truncation, and the finest errors of the seeds sit on it (Section 4.4).

Against each $\delta > 0$ line the figures also draw the *floor* $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$ at the nodes, the distance between the jump problem's solution u_0 and the smooth one u_δ , from the two references. It is what a method that converges to the jump's solution reaches (Section 4.3).

4.2. The knee. Figure 2 gives the three operators at every width, and Table 1 the naive operator's and the construction's errors on the ramp problem. At $\delta = 0$ the naive operator is first order, with rates of 0.98 to 1.02 on the two-constant medium and 1.00 to 1.01 on the sinusoidal one from 201 nodes. While $h \geq 4\delta$ its lines at $\delta > 0$ stay with the jump's, at 0.43 to 1.05 times its error and at rates of 0.9 to 1.6 that depend on where the nodes happen to sample the edge: its rows that see the edge treat it as a jump between two of their nodes, the first-order failure of Section 3.4.

The knee is at $h \approx \delta$. Between $h = 2\delta$ and $h = \delta/2$ the ramp problem's error falls by 117 and 199 times on the two-constant medium at $\delta = 0.01$ and 0.0025, and by 771 and 977 times on the sinusoidal medium, where a first-order line falls by four. On the two-constant medium, whose edge lies midway between two nodes, most of the fall comes in the one doubling from $h = \delta$ to $\delta/2$, at rates of 5.9 to 6.8; on the sinusoidal medium, whose edges lie on nodes, it comes a doubling earlier, at 5.2 to 5.4. From $h = \delta/2$ on, the nodes resolve the tanh and the lines are fourth order, at rates of 3.95

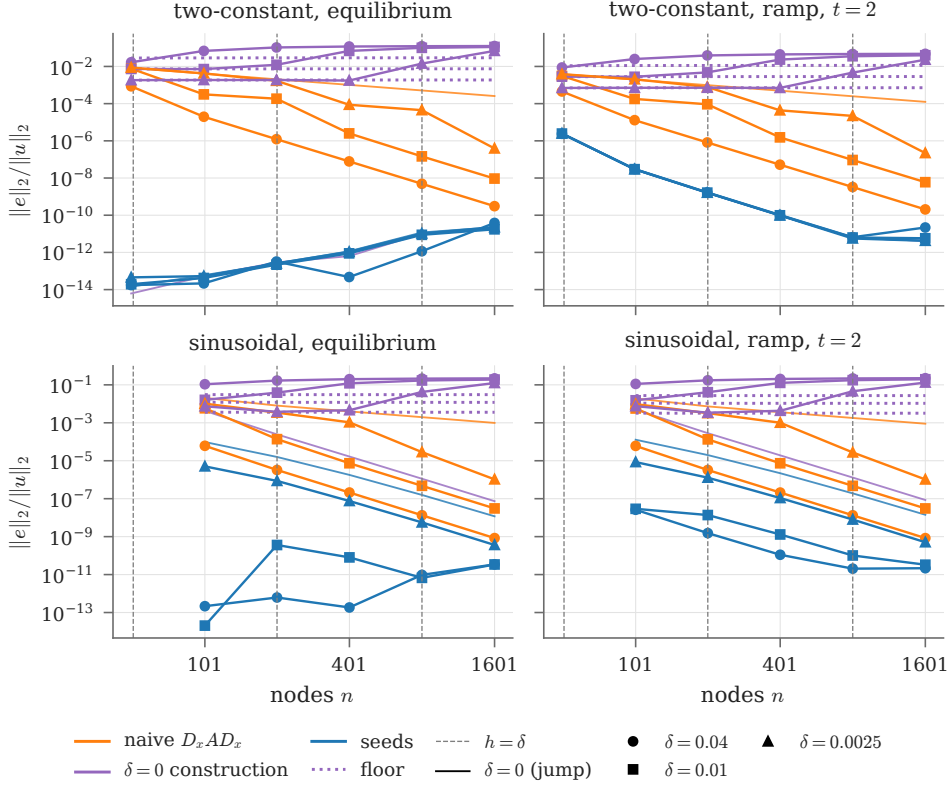


FIGURE 2. The naive operator $D_x A D_x$ (orange), the $\delta = 0$ construction (purple) and the seeds (blue) against the node count n on the two-constant medium (top) and the sinusoidal medium (bottom), at equilibrium (left) and on the ramp problem at $t = 2$ (right); errors are $\|e\|_2/\|u\|_2$ at the nodes. Markers give δ , and lines without markers are the jump, $\delta = 0$. Dotted: the floor $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$ of each $\delta > 0$. Dashed verticals: $h = \delta$. At equilibrium on the two-constant medium the seeds, and at $\delta = 0$ the construction, are exact, and their lines are the direct solve's rounding.

to 4.23 up to the largest count. The equilibrium has the same knee at the same counts: the knee belongs to the operator, not to the problem.

4.3. The jump construction on a smooth edge. The $\delta = 0$ construction converges to the jump problem's solution, so while it converges the floor bounds its error from below. It has two regimes (Figure 2, purple, and Table 1).

Unresolved. While the grid does not resolve the edge, the construction sits on the floor. At $\delta = 0.0025$ on the two-constant medium its equilibrium error at 50, 100 and 200 nodes is the floor to three digits, and wherever $h \geq 2\delta$

TABLE 1. The naive operator and the $\delta = 0$ construction on the ramp problem at $t = 2$: $\|e\|_2/\|u\|_2$ at the nodes, against the Chebyshev reference at each δ . *floor*: the two references' difference $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$ at the largest count. Last line: the constant c of the resistance deficit (5), measured from the two quadratures at every δ and in closed form.

n	naive $D_x A D_x$			$\delta = 0$ construction		
	$\delta = 0.04$	$\delta = 0.01$	$\delta = 0.0025$	$\delta = 0.04$	$\delta = 0.01$	$\delta = 0.0025$
<i>two-constant medium 1/9 1, ramp problem at $t = 2$</i>						
50	4.50×10^{-4}	3.09×10^{-3}	3.85×10^{-3}	8.80×10^{-3}	2.82×10^{-3}	6.97×10^{-4}
100	1.29×10^{-5}	1.79×10^{-4}	2.04×10^{-3}	2.51×10^{-2}	2.78×10^{-3}	7.12×10^{-4}
200	8.19×10^{-7}	9.25×10^{-5}	7.90×10^{-4}	3.91×10^{-2}	4.79×10^{-3}	7.16×10^{-4}
400	5.22×10^{-8}	1.54×10^{-6}	4.36×10^{-5}	4.45×10^{-2}	2.34×10^{-2}	7.01×10^{-4}
800	3.28×10^{-9}	9.32×10^{-8}	2.19×10^{-5}	4.67×10^{-2}	3.54×10^{-2}	4.65×10^{-3}
1600	2.05×10^{-10}	5.96×10^{-9}	2.20×10^{-7}	4.77×10^{-2}	3.99×10^{-2}	2.30×10^{-2}
floor				1.14×10^{-2}	2.87×10^{-3}	7.18×10^{-4}
<i>sinusoidal medium, ramp problem at $t = 2$</i>						
101	6.06×10^{-5}	5.67×10^{-3}	9.61×10^{-3}	1.12×10^{-1}	1.53×10^{-2}	7.51×10^{-3}
201	3.25×10^{-6}	1.34×10^{-4}	3.23×10^{-3}	1.73×10^{-1}	4.09×10^{-2}	3.41×10^{-3}
401	2.11×10^{-7}	7.36×10^{-6}	1.01×10^{-3}	2.04×10^{-1}	1.27×10^{-1}	4.29×10^{-3}
801	1.33×10^{-8}	4.76×10^{-7}	2.74×10^{-5}	2.18×10^{-1}	1.77×10^{-1}	4.60×10^{-2}
1601	8.36×10^{-10}	3.06×10^{-8}	1.03×10^{-6}	2.24×10^{-1}	1.97×10^{-1}	1.32×10^{-1}
floor				2.71×10^{-2}	1.06×10^{-2}	3.20×10^{-3}
two-constant medium: deficit $c\delta$, $c = 8.78890$ measured, 8.78890 closed form						

on that medium it lies at 0.93 to 1.00 times the floor, at equilibrium and on the ramp problem. The floor is first order in δ , 0.73 to 0.75δ at equilibrium and 0.29δ on the ramp problem at $t = 2$, since the edge's resistance falls short of the jump's by $c\delta$ with $c = 8.79$ (5), which the two quadratures reproduce to six digits at every width; the flux through the medium changes by $c\delta/F_0 = 0.88\delta$ relative to the jump's, $F_0 = 10$ being the jump's total resistance. On the sinusoidal medium the floor is not yet proportional to δ at these widths: it is 0.76 , 1.20 and 1.44δ at equilibrium for $\delta = 0.04$, 0.01 and 0.0025 , and the measured deficit is 0.44 , 0.74 and 0.91 of the limit 20.7δ of its two $10 : 1$ edges. The sinusoid has $\alpha'/\alpha = 25$ where it meets the edges, so that at $\delta = 0.01$ it changes by 75% within 3δ of the edge, and the pieces the edge blends are not the constants that (5) assumes. There the coarse rows also carry the sinusoid's own error, and the construction lies at 1.05 to 2.3 times the floor where $h \geq 2\delta$.

Resolved. Once the grid resolves the edge the construction leaves the floor and its error grows. Its plain rows are then fourth-order rows on a smooth coefficient, but its three or four rebuilt rows still impose the translated basis, a kink of slope ratio α^+/α^- at the edge's centre, on a solution whose slope hardly changes across the centre cell, and the discrete solution takes the

TABLE 2. The seeds on the ramp problem at $t = 2$: $\|e\|_2/\|u\|_2$ at the nodes, with the rate per doubling of n in parentheses. Errors of a few times 10^{-12} on the two-constant medium and a few times 10^{-11} on the sinusoidal one are the references' own floor (Section 4.1), where the rates mean nothing.

n	$\delta = 0$	$\delta = 0.04$	$\delta = 0.01$	$\delta = 0.0025$
<i>two-constant medium $1/9 1$, ramp problem, error (rate)</i>				
50	2.49×10^{-6}	2.39×10^{-6}	2.46×10^{-6}	2.48×10^{-6}
100	2.95×10^{-8} (6.40)	3.05×10^{-8} (6.29)	2.99×10^{-8} (6.36)	2.95×10^{-8} (6.39)
200	1.61×10^{-9} (4.19)	1.69×10^{-9} (4.17)	1.64×10^{-9} (4.19)	1.62×10^{-9} (4.19)
400	9.76×10^{-11} (4.05)	1.03×10^{-10} (4.04)	9.89×10^{-11} (4.05)	9.74×10^{-11} (4.05)
800	6.11×10^{-12} (4.00)	6.54×10^{-12} (3.97)	6.32×10^{-12} (3.97)	5.61×10^{-12} (4.12)
1600	4.18×10^{-12} (0.55)	2.18×10^{-11} (-1.74)	5.76×10^{-12} (0.13)	4.22×10^{-12} (0.41)
<i>sinusoidal medium, ramp problem, error (rate)</i>				
101	1.28×10^{-4}	2.61×10^{-8}	2.92×10^{-8}	8.50×10^{-6}
201	1.99×10^{-5} (2.68)	1.53×10^{-9} (4.09)	1.37×10^{-8} (1.09)	1.28×10^{-6} (2.73)
401	2.20×10^{-6} (3.18)	1.11×10^{-10} (3.79)	1.30×10^{-9} (3.40)	1.08×10^{-7} (3.57)
801	1.91×10^{-7} (3.53)	2.05×10^{-11} (2.43)	1.02×10^{-10} (3.68)	7.95×10^{-9} (3.77)
1601	1.42×10^{-8} (3.75)	2.16×10^{-11} (-0.08)	3.31×10^{-11} (1.62)	5.06×10^{-10} (3.97)

kink. On the two-constant medium the error then follows h/δ , not δ : at $h \approx \delta/2$ it is 0.0685 to 0.0687 for all three widths, which is 2.4, 9.2 and 37 times the floor at $\delta = 0.04$, 0.01 and 0.0025. At 1600 nodes it reaches 0.11 at equilibrium for $\delta = 0.01$, and for $\delta = 0.04$ it is 0.125 at equilibrium and 0.048 on the ramp problem (0.22 for both on the sinusoidal medium). Between the two regimes, at $h \approx 2\delta$, the partly resolved plain rows pull the solution toward the smooth one, and the error dips just below the floor, to 0.93 of it at $\delta = 0.0025$.

So treating the edge as a jump is the right baseline exactly where the grid cannot see the edge, and the worst of the three operators where it can. Using it means knowing δ , switching it off near $h = 2\delta$, and accepting a floor that falls only as δ . At $\delta = 0$ it is the interface-aware operator of [Mar16], exact at equilibrium on the two-constant medium and fourth order on the sinusoidal one (rates 3.88 to 4.12).

4.4. The seeds. *The two-constant medium.* Table 2 and the blue lines of Figure 2 give the seeds. On the two-constant medium their error is one line for every δ . At $\delta = 0$ they are the interface-aware operator (Section 3.2). On the ramp problem their errors at $\delta = 0.0025$ lie within 0.1 to 0.3% of the $\delta = 0$ line at every count to 400, at $\delta = 0.01$ within 1 to 1.5%, and at $\delta = 0.04$, where $\delta \approx h$ at 50 nodes and the smooth solution is a different function, within 3 to 5%. The rates at $\delta = 0$ are 6.4, 4.2 and 4.05, then 4.0 to 4.1 at every width, from 2.5×10^{-6} at 50 nodes to 6×10^{-12} at 800, where

every width reaches the references' floor. At equilibrium the seeds are exact at every width (Remark 4.1).

Time and space. These are the errors of the fully discrete scheme at $\Delta t = h$, and part of each is the time integrator's. With $\Delta t = h/8$ the spatial error alone at $\delta = 0$ is 1.47×10^{-8} , 1.05×10^{-9} , 7.40×10^{-11} and 4.5×10^{-12} at 100 to 800 nodes, rates 3.8, 3.8 and 4.0, with the $\delta = 0.0025$ line within 0.3% of it and the $\delta = 0.01$ line within 1.8%. The fourth order and the one line are the space discretisation's. The time error is about half the total at 100 nodes and a quarter from 200, and it is where the first rate of 6.4 comes from.

A resolved edge costs nothing. Across the knee, where the naive operator changes order and the construction leaves its floor, the seeds' line does not move: it is the same fourth-order line at every h/δ from 16 down to $1/8$, below which the references' floor takes over, with no threshold in δ (Section 3.1). At 200 nodes and $\delta = 0.0025$, where $h = 4\delta$, the seeds' error on the ramp problem is 1.6×10^{-9} , against the naive operator's 7.9×10^{-4} and the construction's 7.2×10^{-4} . The construction's floor falls as δ , so the seeds' margin over it is set by δ and closes at $\delta = 0$, where on constant pieces the two are one operator.

The sinusoidal medium. At equilibrium the seeded rows are exact on the sinusoidal medium too, and the seed operator is exact only where the seeded rows cover the layer. On both problems the rows whose stencils lie farther than 20δ from both edges are standard rows on the sinusoid, and their error is the line. At $\delta = 0.04$ the reach covers the layer, every row is seeded, and the ramp problem's error is 2.6×10^{-8} , 1.5×10^{-9} and 1.1×10^{-10} at 101, 201 and 401 nodes (rates 4.1 and 3.8), and then the references' floor. At $\delta = 0.01$ the reach covers the layer at 101 nodes only; from 201 on it leaves the rows near $x = 0.25$ standard, which carry 1.4×10^{-8} at 201 nodes, and the line is fourth order past them. With every row seeded instead, the $\delta = 0.01$ errors at 201 nodes are 1.3×10^{-13} at equilibrium and 1.7×10^{-9} on the ramp problem; the reach is the operator's one parameter, and we left it at 20δ . At $\delta = 0.0025$ and at the jump the interior of the layer is standard at every count, and the line is the sinusoid's own error under the standard fourth-order rows, which is still pre-asymptotic, the rows beside the edges seeing $h\alpha'/\alpha = 0.5$ at 101 nodes. At $\delta = 0$ it falls from 1.3×10^{-4} to 1.4×10^{-8} over 101 to 1601 nodes at rates rising from 2.7 to 3.75, and at $\delta = 0.0025$ its rates rise from 2.7 to 4.0. At $\delta = 0$ that line is still 6 to 38 times below the construction's (Section 3.2).

Stability. The seed operator's interior spectrum, with the Dirichlet rows removed as BD4 steps it, is real and negative on both media at $\delta = 0$ and 0.0025, and its extreme eigenvalue is the standard stencil's $-16/(3h^2)$ in the material with $\alpha = 1$, to 0.6% at 50 nodes. On the sinusoidal medium at 49 and 53 nodes, where $h\alpha'/\alpha \approx 1$ at the layer's edges, the seed operator is stable, with largest eigenvalue -2.05 , while the $\delta = 0$ construction has

eigenvalues at +248 and +8.8; the sweep starts at 101 nodes so that all three operators share its grids.

4.5. Coefficient treatments. The standing alternative to a new stencil changes the coefficient and keeps the scheme (Section 1.1). We ran four such treatments on the grids and problems above, against the same references:

- T1:** the harmonic mean $(b - a) / \int_a^b d\xi / \alpha$ of α over a window $[a, b]$ of one or two cells centred on each node, clipped at the ends of the domain and sampled into A in $D_x A D_x$;
- T2:** the arithmetic mean over one cell, used the same way;
- T0:** the edge widened to $\max(\delta, mh)$, $m = 1$ or 2 , a coefficient regularised on the scale of the grid [TE03];
- T1-FV:** the conservative three-point scheme with the face conductances (13).

T1-FV is Pan, Xu and Li’s harmonic average method [PXL26], exact at the one-dimensional equilibrium (Section 3.5). T1 is the like-for-like changed medium: a harmonic cell mean, which as a face conductance makes the conservative scheme exact at equilibrium, sampled at the nodes of the fourth-order operator, where it does not; we found no source that uses it there. The harmonic mean is the exact conductance between blocks of uniform conductivity, and may be inferior to other means where the conductivity varies smoothly between nodes [GA92], as it does here once $\delta > 0$. A fifth candidate, the band-limited coefficient of the seismic literature [Mit17, KWR22], was not found in use for diffusion and is left out. The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension.

The ranking. Figure 3 gives every treatment on the ramp problem, and Table 3 the errors at $\delta = 0.0025$ in the order of the ranking. On the ramp problem of the two-constant medium, at every width while $h \geq 4\delta$, the jump included, the order in $\|e\|_2 / \|u\|_2$ is

$$\begin{aligned} \text{seeds} < \text{T1-FV} < \text{T1 (two cells)} < \text{naive} \gtrsim \text{T1 (one cell)} \\ &\approx \text{T2} < \text{T0} (m = 1) < \text{T0} (m = 2), \end{aligned}$$

with the middle three at most 2.02 times apart; at 200 nodes and $\delta = 0.0025$ the errors are 1.6×10^{-9} , 8.0×10^{-6} , 8.9×10^{-5} , 7.9×10^{-4} , 4.0×10^{-4} , 4.6×10^{-4} , 2.2×10^{-3} and 5.0×10^{-3} . At the jump the seeds lead T1-FV by 53 times at 50 nodes and by 2.1×10^4 at 400. On the sinusoidal medium, whose edges fall on nodes, a one-cell window straddles each edge, and at the jump the one-cell harmonic mean is 9 to 35 times below the naive operator. There T1-FV is ahead of the seeds at $\delta = 0$ below 801 nodes and at $\delta = 0.0025$ at 101, where the seeds’ line is the sinusoid’s pre-asymptotic

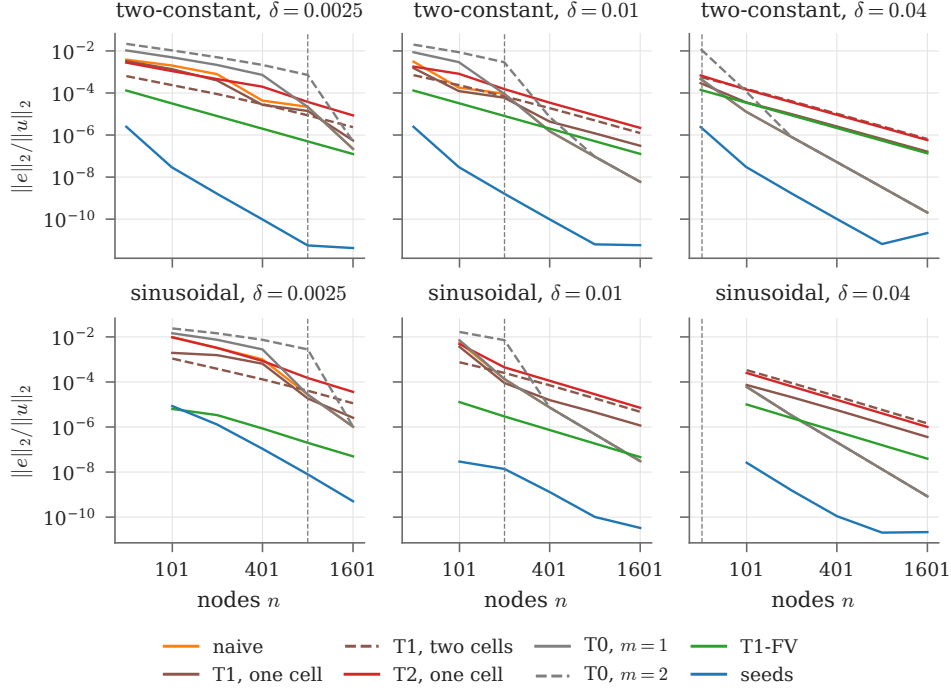


FIGURE 3. The coefficient treatments on the ramp problem at $t = 2$: $\|e\|_2/\|u\|_2$ at the nodes against n on the two-constant medium (top) and the sinusoidal medium (bottom) at $\delta = 0.0025, 0.01$ and 0.04 . The naive operator and the seeds are those of Figure 2, and the treatments those of Table 3. Dashed verticals: $h = \delta$. The seeds' flat ends are the references' floor.

one (Section 4.4) rather than the edge's; where the seeded rows cover the layer the seeds lead, by 440 times at 101 nodes and $\delta = 0.01$.

Second order. T1-FV is second order at every width, as Pan, Xu and Li prove, and on the two-constant medium with one constant: at $\delta = 0$ its error falls from 1.31×10^{-4} at 50 nodes to 1.26×10^{-7} at 1600, rate 2.00; at $\delta = 0.01$ and 0.0025 its errors lie within 1.2% of these at every count, and at $\delta = 0.04$ they lie 7% above. It is the strongest of the treatments. Every nodal treatment caps the fourth-order operator at second order once the grid resolves the edge: for $h \leq \delta/4$ on the two-constant medium the naive operator is fourth order and the treated ones second order (rates 1.95 to 2.00). At $\delta = 0.04$ and 1600 nodes the naive error is 2.05×10^{-10} , and the treated errors are 1.62×10^{-7} (T1, one cell), 5.76×10^{-7} (T2) and 6.50×10^{-7} (T1, two cells), 1.2 to 4.8 times T1-FV's 1.35×10^{-7} . A cell mean perturbs a smooth α at second order in the width of its window, and the fourth-order operator solves the perturbed problem faithfully. For the wave

TABLE 3. The treatments on the ramp problem at $t = 2$ and $\delta = 0.0025$: $\|e\|_2/\|u\|_2$ at the nodes, to two figures, in the order of the ranking of Section 4.5. T1: the harmonic mean over one or two cells; T2: the arithmetic mean over one cell; T0: the edge widened to $\max(\delta, mh)$; T1-FV: the conservative scheme with the face conductances (13).

$\ e\ _2/\ u\ _2$						
<i>two-constant medium 1/9 1, ramp problem, $\delta = 0.0025$</i>						
n	50	100	200	400	800	1600
seeds	2.5×10^{-6}	3.0×10^{-8}	1.6×10^{-9}	9.7×10^{-11}	5.6×10^{-12}	4.2×10^{-12}
T1-FV	1.3×10^{-4}	3.2×10^{-5}	8.0×10^{-6}	2.0×10^{-6}	5.0×10^{-7}	1.3×10^{-7}
T1, two cells	6.4×10^{-4}	2.4×10^{-4}	8.9×10^{-5}	3.0×10^{-5}	8.7×10^{-6}	2.4×10^{-6}
naive	3.8×10^{-3}	2.0×10^{-3}	7.9×10^{-4}	4.4×10^{-5}	2.2×10^{-5}	2.2×10^{-7}
T1, one cell	3.3×10^{-3}	1.4×10^{-3}	4.0×10^{-4}	2.7×10^{-5}	1.4×10^{-5}	5.5×10^{-7}
T2, one cell	2.8×10^{-3}	1.1×10^{-3}	4.6×10^{-4}	2.0×10^{-4}	3.8×10^{-5}	8.6×10^{-6}
T0, $m = 1$	1.1×10^{-2}	5.0×10^{-3}	2.2×10^{-3}	7.2×10^{-4}	2.2×10^{-5}	2.2×10^{-7}
T0, $m = 2$	2.2×10^{-2}	1.1×10^{-2}	5.0×10^{-3}	2.2×10^{-3}	7.2×10^{-4}	5.0×10^{-7}
<i>sinusoidal medium, ramp problem, $\delta = 0.0025$</i>						
n	101	201	401	801	1601	
seeds	8.5×10^{-6}	1.3×10^{-6}	1.1×10^{-7}	8.0×10^{-9}	5.1×10^{-10}	
T1-FV	6.4×10^{-6}	3.4×10^{-6}	8.6×10^{-7}	2.0×10^{-7}	5.0×10^{-8}	
T1, two cells	1.1×10^{-3}	3.8×10^{-4}	1.3×10^{-4}	4.1×10^{-5}	1.1×10^{-5}	
naive	9.6×10^{-3}	3.2×10^{-3}	1.0×10^{-3}	2.7×10^{-5}	1.0×10^{-6}	
T1, one cell	2.0×10^{-3}	1.6×10^{-3}	6.5×10^{-4}	1.9×10^{-5}	2.6×10^{-6}	
T2, one cell	9.7×10^{-3}	3.2×10^{-3}	8.6×10^{-4}	1.5×10^{-4}	3.6×10^{-5}	
T0, $m = 1$	1.5×10^{-2}	7.4×10^{-3}	2.8×10^{-3}	2.7×10^{-5}	1.0×10^{-6}	
T0, $m = 2$	2.4×10^{-2}	1.4×10^{-2}	7.4×10^{-3}	2.8×10^{-3}	1.0×10^{-6}	

equation with regularised discontinuities, Tornberg and Engquist report the same split: a second-order error component from the discontinuities under a higher-order discretisation [TE06]. So the nodal treatments share the construction's property: they help only while $h \gtrsim \delta$, and using them means knowing δ and switching them off, which the seeds need not.

The widened edge and the local defect. T0 is the worst. The widened medium's equilibrium differs from the true one by a resistance deficit $c(mh - \delta)$, and T0's equilibrium error is that difference, to 0.1% for $m = 2$ wherever $h \geq \delta$; on the two-constant medium's ramp problem at $\delta = 0$ it is 2.9 ($m = 1$) and 5.7 to 5.8 ($m = 2$) times the naive operator's error at every count. Widening an edge that the grid already fails to resolve adds resistance error and leaves the knee where it was. With the edge midway between nodes, a one-cell window ends on the jump at $\delta = 0$, and T1 (one cell) and T2 are the naive operator there to rounding. T1 over two cells converges at order 1.5 in this norm (rates 1.44 to 1.50 at $\delta = 0$ on the two-constant medium's ramp problem), the mark of a local defect: its error is first order on the two nodes beside the edge, 1.18×10^{-3} to 7.28×10^{-5} in the maximum norm

TABLE 4. The elliptic problem: $\|e\|_2/\|u\|_2$ at the nodes for the naive operator and the $\delta = 0$ construction at $\delta = 0.01$, the seeds at three widths, and T1-FV at $\delta = 0$ (Remark 4.1).

n	$\delta = 0.01$		seeds			T1-FV
	naive	construction	$\delta = 0$	$\delta = 0.01$	$\delta = 0.0025$	$\delta = 0$
<i>two-constant medium</i> $1/9 \mid 1$, <i>equilibrium</i> , $\ e\ _2/\ u\ _2$						
50	7.64×10^{-3}	7.35×10^{-3}	1.4×10^{-14}	1.9×10^{-14}	4.5×10^{-14}	2.7×10^{-14}
100	3.15×10^{-4}	7.21×10^{-3}	5.9×10^{-14}	4.2×10^{-14}	5.3×10^{-14}	8.2×10^{-15}
200	1.84×10^{-4}	1.23×10^{-2}	2.6×10^{-13}	2.2×10^{-13}	2.6×10^{-13}	1.2×10^{-13}
400	2.51×10^{-6}	6.86×10^{-2}	1.2×10^{-12}	8.9×10^{-13}	1.1×10^{-12}	2.1×10^{-13}
800	1.46×10^{-7}	1.00×10^{-1}	1.2×10^{-11}	8.8×10^{-12}	1.0×10^{-11}	1.7×10^{-12}
1600	9.34×10^{-9}	1.12×10^{-1}	2.3×10^{-11}	1.7×10^{-11}	2.3×10^{-11}	6.5×10^{-12}
<i>sinusoidal medium</i> , <i>equilibrium</i> , $\ e\ _2/\ u\ _2$						
101	5.83×10^{-3}	1.62×10^{-2}	9.7×10^{-5}	2.0×10^{-14}	5.0×10^{-6}	3.0×10^{-14}
201	1.38×10^{-4}	3.89×10^{-2}	1.6×10^{-5}	3.6×10^{-10}	8.6×10^{-7}	1.3×10^{-13}
401	7.38×10^{-6}	1.21×10^{-1}	1.8×10^{-6}	8.0×10^{-11}	7.5×10^{-8}	1.6×10^{-13}
801	4.77×10^{-7}	1.70×10^{-1}	1.6×10^{-7}	6.7×10^{-12}	5.6×10^{-9}	2.2×10^{-12}
1601	3.07×10^{-8}	1.90×10^{-1}	1.2×10^{-8}	3.4×10^{-11}	3.6×10^{-10}	1.4×10^{-11}

from 100 to 1600 nodes, and far smaller elsewhere, so that the relative ℓ^2 error goes as $h n^{-1/2}$. In the maximum norm it is first order.

Remark 4.1 (The elliptic problem). The left column of Figure 2 and Table 4 are the sanity check and the limit case, not a ranking (Section 3.5). On the two-constant medium the seeds are exact at every width, 1.9×10^{-14} at 50 nodes and $\delta = 0.01$, growing with n to 1.7×10^{-11} at 1600 through the direct solve's conditioning, as the interface-aware operator's own line at $\delta = 0$ does; T1-FV is exact too, 10^{-14} to 6.5×10^{-12} . The naive operator shows the knee of the ramp problem, from 7.6×10^{-3} at $h = 4\delta$ to 9.3×10^{-9} at $h = \delta/8$ for $\delta = 0.01$, and the construction the same floor. On the sinusoidal medium the equilibrium does rank the two, for a reason that has nothing to do with the edge. T1-FV stays exact, 3×10^{-14} to 1.4×10^{-11} at $\delta = 0$, while the seed operator is exact only where its seeded rows cover the layer, at $\delta = 0.04$ and, at 101 nodes, $\delta = 0.01$; elsewhere its error is that of the standard rows inside the layer, 9.7×10^{-5} to 1.2×10^{-8} at $\delta = 0$ over 101 to 1601 nodes and 5.0×10^{-6} to 3.6×10^{-10} at $\delta = 0.0025$.

4.6. A snapshot. Figure 4 and Table 5 put the four operators on one grid: the ramp problem at $t = 2$ on 100 nodes of the two-constant medium with $\delta = 0.0025$, so that $h = 8.08\delta$. It is the row of the sweep at which the four lines are ordered and apart: the naive operator at 2.04×10^{-3} , the construction 2.9 times below it, T1-FV 22 times below that, and the seeds 1100 times below that, at 2.95×10^{-8} . The naive operator's error is largest beside the edge, $1.5h$ to its left, and carries a node-to-node oscillation, the checkerboard mode of $D_x A D_x$ excited by the sampled edge; but 84% of $\|e\|_2^2$ lies more than $2h$ from the edge, on the side where $\alpha = 1/9$ and the profile

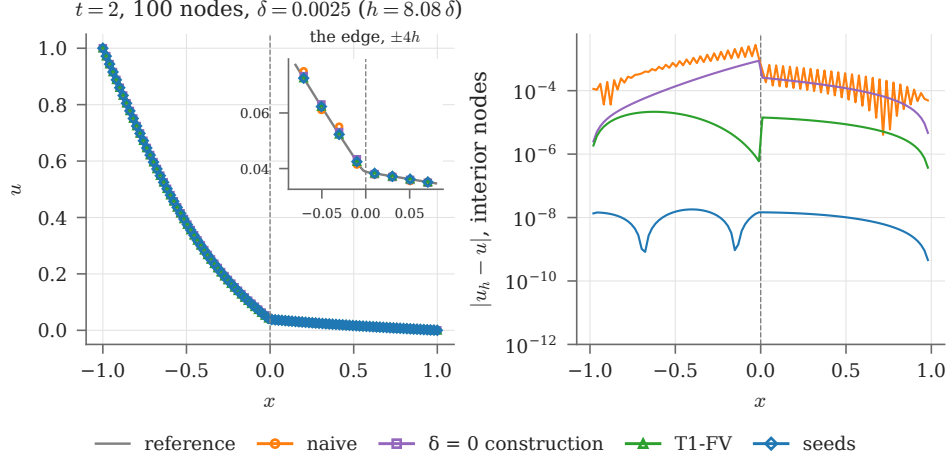


FIGURE 4. The ramp problem at $t = 2$ on 100 nodes of the two-constant medium with $\delta = 0.0025$ ($h = 8.08\delta$). Left: the reference and the nodal solutions of the four operators, with the edge enlarged over $\pm 4h$. Right: the pointwise error $|u_h - u|$ at the interior nodes. Dashed: the edge's centre.

TABLE 5. The snapshot of Figure 4: $\|e\|_2/\|u\|_2$ at the nodes, the largest nodal error and where it lies, and the share of $\|e\|_2^2$ from the nodes within $2h$ of the edge.

operator	$\ e\ _2/\ u\ _2$	$\max e $	at x	share within $2h$
naive $D_x A D_x$	2.04×10^{-3}	2.78×10^{-3}	-0.030	0.16
$\delta = 0$ construction	7.12×10^{-4}	8.79×10^{-4}	-0.010	0.23
T1-FV	3.20×10^{-5}	2.16×10^{-5}	-0.616	0.03
seeds	2.95×10^{-8}	1.80×10^{-8}	-0.394	0.07

is steep. The construction's error is smooth and global, the profile shifted by the resistance deficit, and largest at the node half a cell from the edge. T1-FV's is largest far from the edge, at $x = -0.62$: the face conductances take the edge exactly, and what remains is second-order truncation on the curved profile. The seeds' error is the $\delta = 0$ line's at this count, flat across the domain but for two sign changes, and about half of it is BD4's: with $\Delta t = h/8$ it is 1.47×10^{-8} .

5. SEEDS IN TWO DIMENSIONS

A stencil of the interface group (Section 2.3) has 30 nodes and degree four, and the jump-aware construction replaces its fifteen monomials by the translated basis. Here they are replaced by seeds, the profiles at $t = 0$ of solutions polynomial in time, anchored at the evaluation node as in Section 3 and now functions of two variables. The node sets, the stencil

sizes, the Gaussians and the saddle-point solve are kept. So is the rest of the operator: the bulk rows, the Dirichlet rows and the time stepping. The construction for a straight edge is the companion paper's for the wave equation [Mar26], specialised to one field (Sections 5.1 and 5.2); a curved or tangentially varying edge and a resistive layer thinner than the stencil each add to it (Sections 5.3 and 5.4).

5.1. A straight feature. *Frame and anchor.* A seeded stencil keeps the orientation of the jump-aware frame: at the foot point of the interface nearest the evaluation node, x' along the tangent and y' along the normal (Section 2.4). Its origin moves to the evaluation node x_e , which lies on the normal through that foot point, and lengths are measured in the stencil's radius h_s , the largest distance from x_e to a node of the stencil:

$$(14) \quad \xi = (x' - x'_e)/h_s, \quad \eta = (y' - y'_e)/h_s.$$

The material along the normal line, $\alpha_n(\eta) = \alpha(x_e + h_s\eta n)$ with n the unit normal, is what the seeds read, and $\alpha_e = \alpha_n(0)$. In these coordinates $L = h_s^{-2}(\partial_\xi \alpha \partial_\xi + \partial_\eta \alpha \partial_\eta)$; as in Section 3.1 we drop the factor h_s^{-2} in what follows and restore it in the weights, $w = \tilde{w}/h_s^2$.

The ansatz. Where α depends on the normal coordinate alone, as it does exactly across the flat edges of case 1, the seed of the monomial $\xi^a \eta^b$ is a polynomial in ξ whose coefficients are functions of η ,

$$(15) \quad \phi_{ab}(\xi, \eta) = \sum_j g_j(\eta) \xi^j, \quad j = a, a-2, \dots, a \bmod 2,$$

because $L(g(\eta)\xi^j) = \xi^j(\alpha_n g')' + j(j-1)\alpha_n g \xi^{j-2}$: the operator lowers the degree in ξ by two or not at all. The chain (2) read in two dimensions is

$$(16) \quad \phi_{00} = 1, \quad L\phi_{ab} = \alpha_e[a(a-1)\phi_{a-2,b} + b(b-1)\phi_{a,b-2}],$$

the constant-coefficient operator's action on the monomial with the lower monomials' seeds on the right. Collecting ξ^j and writing the flux $\psi_j = \alpha_n g'_j$ gives a first-order pair for each seed and level,

$$(17) \quad \begin{aligned} g'_j &= \psi_j / \alpha_n, \\ \psi'_j &= \alpha_e[a(a-1)g_j^{(a-2,b)} + b(b-1)g_j^{(a,b-2)}] - (j+2)(j+1)\alpha_n g_{j+2}^{(a,b)}, \end{aligned}$$

where a superscript names the seed a level belongs to and a missing level is zero. The initial data are the monomial's at its top level, $g_a(0) = [b=0]$ and $\psi_a(0) = \alpha_e[b=1]$, and zero for every other level; for $b \geq 2$ the source generates η^b , as $\phi_k(x_e) = \phi'_k(x_e) = 0$ did in one dimension. Seed (a, b) at level j needs the same level of the seeds $(a-2, b)$ and $(a, b-2)$ and level $j+2$ of itself, so the system is triangular in total degree and in level, and the fifteen seeds of degree at most four are 22 levels with 44 states, marched together as one linear system that never differentiates α . With a constant α the seeds are the monomials, by induction on the degree as in Lemma 3.2. The proof of that lemma, applied once in each index, shows

that $u_{ab} = \sum_{i \leq a/2, k \leq b/2} c_i^a c_k^b (\alpha_e t)^{i+k} \phi_{a-2i, b-2k}$, with $c_i^a = a!/(i!(a-2i)!)$, solves the heat equation in these coordinates with $u_{ab}(\cdot, 0) = \phi_{ab}$: the seeds are again the profiles of solutions polynomial in time.

Closed forms. The linear seeds are the one-dimensional ones along the normal, stretched along the tangent: $\phi_{10} = \xi$, $\phi_{01} = \alpha_e \int_0^\eta d\eta'/\alpha_n$, the constant-flux seed ϕ_1 of Section 3.1, and $\phi_{11} = \xi\phi_{01}$; ϕ_{02} is the seed ϕ_2 along the normal. The first seed that is not a one-dimensional seed times a power of ξ is

$$(18) \quad \phi_{20} = \xi^2 + g_0(\eta), \quad (\alpha_n g_0')' = 2(\alpha_e - \alpha_n), \quad g_0(0) = g_0'(0) = 0.$$

$L\xi^2 = 2\alpha_n$ is not constant across the edge, and g_0 is the normal profile that restores $L\phi_{20} = 2\alpha_e$, the constant rate of change of a temperature quadratic along the edge. It vanishes on a constant α and is of order one in stencil units across an unresolved edge (Figure 5), and no one-dimensional construction supplies it: the two-dimensional content of the seeds is this coupling of tangential monomials to normal profiles. It is also why the elliptic problem tests them in two dimensions where it could not in one (Section 3.5): case 1's equilibrium $\sin 2\pi x v(y)$ is not in the span of the one-dimensional seeds at any resolution.

A symmetry, and the march. The frozen problem is invariant under shifts along ξ , and matching powers of the shift gives $g_j^{(a,b)} = \binom{a}{j} g_0^{(a-j,b)}$, with $g_0^{(m,b)} = 0$ for odd m : only nine level-zero functions are independent. The 44-state march reproduces the identity to 7×10^{-16} at every δ tested, from the jump to $8h$, and the monomials on a constant α to 1.4×10^{-15} ; we march all 44 states because it is the coupled form that carries over to a curved or tangentially varying edge (Section 5.3). The march is that of Section 3.1: SciPy's DOP853 at the same tolerances in the stencil coordinate, from the anchor in both directions, restarted at every node's η so that node values are integrated and not interpolated, with segment boundaries at each edge centre and at 10δ on either side; α on a segment is read from the piece that owns it, so a jump is marched like any other width, and a band thinner than the stencil is crossed in one march with both of its edges as boundaries. The true operator applied to the marched seeds by twelfth-order differences on a 25×801 grid leaves $L\phi_{ab}$ minus the right-hand side of (16) below 10^{-10} relative, the floor of the differences.

5.2. Stencil assembly. *The saddle-point system.* The seeds take the monomials' place in (6): $P_{ie} = \phi_e(\xi_i, \eta_i)$, read off the march at the nodes, and the right-hand side is the true operator on each seed at the anchor,

$$(19) \quad (L\phi_{ab})(x_e) = 2\alpha_e [(a, b) \in \{(2, 0), (0, 2)\}] + h_s \alpha_\xi [(a, b) = (1, 0)],$$

with α_ξ the tangential derivative of α at the anchor. The chain gives the first term, since the right-hand side of (16) vanishes at the anchor for every seed but the two quadratics, whose lower seed is $\phi_{00} = 1$. The second is what the frozen normal profile leaves out: $\partial_\xi(\alpha \partial_\xi \phi) = \alpha_\xi \phi_\xi + \alpha \phi_{\xi\xi}$, and

at the anchor $\phi_\xi \neq 0$ only for $\phi_{10} = \xi$. The normal derivative of α never appears, since the seeds satisfy $\partial_\eta \alpha_n \partial_\eta$ with the true profile. Where the jump-aware rows apply $\alpha \nabla^2 + \nabla \alpha \cdot \nabla$ to all fifteen functions of their basis, the seeds carry $\nabla \alpha \cdot \nabla$ through the march and need it at the anchor once, in one entry.

The warp is the constant-flux seed. The jump-aware stencils warp the Gaussians' normal coordinate piecewise-linearly, with its slope changed by the factor α^-/α^+ at the interface (Section 2.4). The seeds bring the smooth version with them: the Gaussians are written in $(\xi, \tilde{\eta})$ with

$$(20) \quad \tilde{\eta} = \phi_{01}(\eta) = \alpha_e \int_0^\eta \frac{d\eta'}{\alpha_n(\eta')},$$

read off the same march at every node. With $\tilde{\eta}' = \alpha_e/\alpha_n$ the chain rule gives, at any point of the normal line,

$$L G(\xi, \tilde{\eta}(\eta)) = \alpha_\xi G_\xi + \alpha G_{\xi\xi} + (\alpha \tilde{\eta}')' G_{\tilde{\eta}} + \alpha \tilde{\eta}'^2 G_{\tilde{\eta}\tilde{\eta}},$$

and $\alpha \tilde{\eta}' = \alpha_e$ is constant, the flux of ϕ_{01} , so the term in $G_{\tilde{\eta}}$, which the jump-aware rows carry as $\alpha_\eta G_\eta$, cancels exactly. At the anchor

$$(21) \quad (LG)(x_e) = \alpha_e (G_{\xi\xi} + G_{\tilde{\eta}\tilde{\eta}}) + \alpha_\xi G_\xi,$$

the Gaussian's Laplacian at the warped offsets times α_e , with the seeds' tangential correction. At $\delta = 0$ on constant pieces (20) is the jump-aware warp, to 1.8×10^{-15} at the nodes of case 1's 576 crossing stencils at 2500 nodes. The implementation assembles the general chain-rule form and checks the cancellation along the whole marched line rather than assuming it, and takes ε from the physical offsets, as the jump-aware stencils do. Written in the frame's own (ξ, η) the Gaussians are plain and carry $\alpha_\eta G_\eta$; Section 6.4 measures what the warp is worth.

The jump as the limit. At $\delta = 0$ on constant pieces across a flat interface, the seed of $\xi^a \eta^b$ is the monomial on the anchor's side and, on the far side, a polynomial of degree at most four, since the chain has constant coefficients and polynomial sources there. It is continuous with its flux at every ξ , and by induction down the chain so are $L^k \phi$ and their fluxes: the fifteen conditions of the continuity matrices, all exact for constant pieces on a flat line (Section 2.4). The matrices being square and nonsingular, the far side is the translated polynomial, and the seeds and the translated basis are two bases of one space, anchored at the node and at the foot point. With the same Gaussians the seed rows are therefore the jump-aware rows, the two-dimensional form of the one construction with two implementations of Section 3.2. Over the 576 crossing stencils of case 1 at 2500 nodes the sine of the largest principal angle between the two spans is 4.9×10^{-14} , and the weights differ by 1.6×10^{-12} relative with plain Gaussians and by 7.6×10^{-12} with warped ones; assembled, the two operators agree to 5.3×10^{-13} . On a band one spacing thick, 333 crossing stencils of which 240 reach all three regions, where the jump-aware construction translates twice with a change

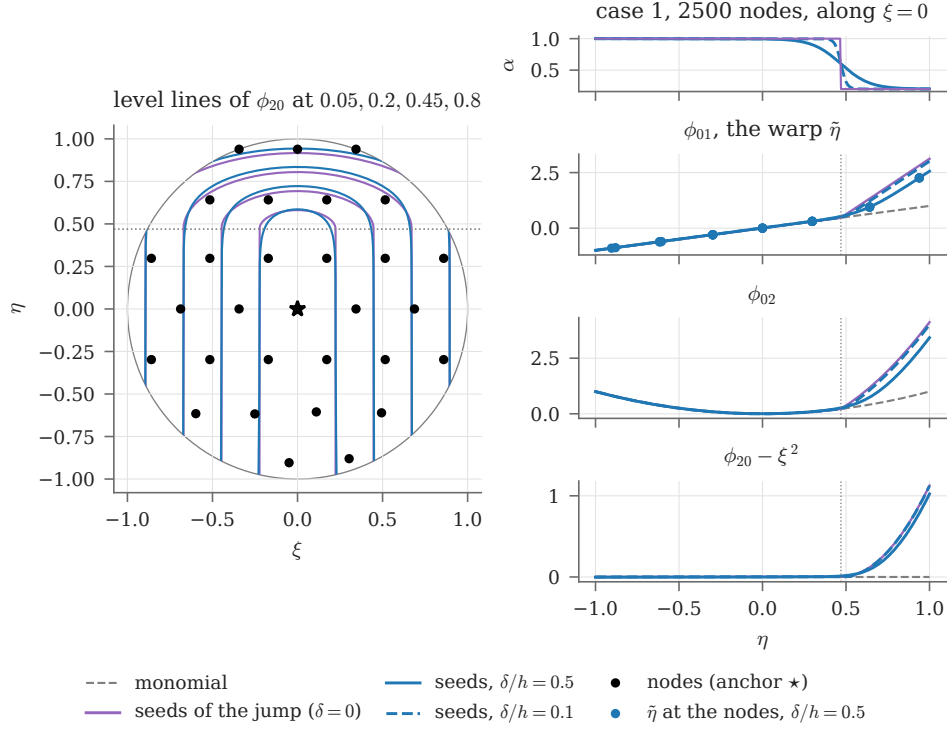


FIGURE 5. The seeds of one 30-node stencil of case 1 at 2500 nodes, anchored (star) on the second straddling row below the lower edge (dotted), in the stencil coordinates (14); the edge is at $\eta = 0.47$ and 7 of the nodes lie across it. Left: the nodes and the level lines of ϕ_{20} for the jump (purple) and for $\delta = h/2$ (blue). Below the edge they are those of ξ^2 ; above it they close, since g_0 adds the normal curvature that keeps $L\phi_{20} = 2\alpha_e$ where $\alpha = 0.2$. Right: α and three seeds along the normal line $\xi = 0$ against the monomials (grey, dashed): the constant-flux seed ϕ_{01} , which is the Gaussians' normal coordinate $\tilde{\eta}$ (dots: its values at the nodes for $\delta = h/2$), ϕ_{02} , and $g_0 = \phi_{20} - \xi^2$, which vanishes on a constant α . Purple: the jump's seeds, which span the translated basis of the jump-aware stencil; blue: $\delta = h/2$ (solid) and $h/10$ (dashed). The anchor's α_e is within 0.4% of the jump's at $\delta = h/2$, so the seeds are drawn as marched, unnormalised.

of frame between the interfaces and the march crosses both in one pass, the span distance is 1.7×10^{-14} and the weights differ by 6.0×10^{-13} . At $\delta > 0$ the seed span leaves the jump's at first order in δ/h , 0.46 to $0.74\delta/h$ across four anchors, with no floor down to $\delta = 10^{-5}h$. On a curved interface, or

with pieces that vary along it, the two constructions differ at $O(h)$, and Section 6 measures each against the reference.

Which rows are seeded. A row is seeded when the signed distances of its 30 nodes to an edge's centre curve span a range that meets $[-20\delta, 20\delta]$, beyond which the edge is its pieces to the last bit (Section 2.1), or when its nodes straddle the curve; at $\delta = 0$ the rule is the jump-aware construction's crossing test. The stencil groups follow the same rule one size up: a node whose 42-node stencil sees an edge joins the interface group and gets a 30-node stencil of degree four, so that no bulk stencil sees an unresolved edge, and of the group the members whose own 30 nodes see it are seeded. A stencil of the boundary zone that sees an edge is seeded on its own one-sided nodes, since the chain is anchored at the node and marched only where nodes are. Once 20δ covers the strip every row is seeded. No threshold in δ switches the seeds off: a resolved edge costs them nothing but the marches (Section 6.4).

Stencil size, conditioning and cost. The seed stencils are the interface group's, 30 nodes and degree four. Our companion paper found degree-four seed stencils unstable across a sharp edge in the wave equation and used 19 nodes and degree three [Mar26]; the cause there was the footprint of the hyperviscosity row the wave operator needs, which has no counterpart in an implicit march of a dissipative operator, and the spectra of Section 6.3 show no eigenvalue in the right half-plane. Column-scaled, the seed block's condition number stays within 0.46 to 2.5 times the monomial block's on the same nodes at every δ from $10^{-5}h$ to $8h$; unscaled, it carries the contrast, from 55 with the edge resolved to 212 with it unresolved on the stencil below the band. A seed row costs 1.2 to 5 times a jump-aware row, a few milliseconds.

5.3. Curved and tangentially varying features. *A frozen profile.* For a curved edge the natural first step, the companion paper's for the wave equation [Mar26], marches the straight-feature seeds along the true normal through the foot point and drops the curvature. The seeds' kink then lies on the tangent line instead of on the curve: a node at tangential offset x' is at normal distance $y' - \kappa x'^2/2 + \dots$ from a curve of curvature κ , while its seed value assumes y' . Moving an unresolved edge by Δ changes the values beyond it by Δ times the jump in the slope that carries the flux, whatever δ is: an error of $O(\kappa h^2)$ in value and, in a row whose weights are $O(h^{-2})$, a truncation error of $O(\kappa)$ that does not converge. At $\delta = 0$ on constant pieces this frozen profile is the jump-aware construction without its curvature terms, whose flat line in [MF17b, Fig. 10] is first order. A variation of α along an unresolved edge does the same another way: the ratio of fluxes the seeds impose across it changes from node to node, a kinked correction outside their span. Either term alone makes a frozen profile inconsistent at $O(1)$ through an unresolved edge; Section 6.5 measures the two apart.

The curve's own coordinates. Write every point near the foot curve $\gamma(\sigma)$, in its own parameter, as $x = \gamma(\sigma) + d n(\sigma)$, with n the unit normal and κ the signed curvature. These coordinates are one-to-one inside the focal distance, and the implementation refuses a stencil whose normal line reaches within a margin of it, $\kappa_{\max}(|d_e| + 2h_s) \geq 0.8$, with $\kappa_{\max}$ the curve's largest curvature and d_e the anchor's distance to it. The metric is diagonal, with scale factors $m = |\gamma'|/(1 - \kappa d)$ along σ and 1 along d , so that

$$(22) \quad Lu = \frac{1}{m} \left[\partial_\sigma \left(\frac{\alpha}{m} \partial_\sigma u \right) + \partial_d (\alpha m \partial_d u) \right],$$

in divergence form with no cross term. About the anchor (σ_0, d_e) the stencil coordinates are $\xi = \mu_e(\sigma - \sigma_0)/h_s$ and $\eta = (d - d_e)/h_s$, with $\mu_e = m(\sigma_0, d_e)$; the operator keeps its form with $\hat{m} = m/\mu_e$ in place of m , and $\hat{m} = 1$ at the anchor, so that ξ is arclength along the anchor's parallel curve in stencil radii. The normal line $\xi = 0$ is the frozen profile's, with the same α_n , segment boundaries and pieces. What changes is that the foot curve is the line $\eta = -d_e/h_s$ at every ξ , so the edge is a coordinate line and its conditions hold along one.

Coupled levels. Keep the ansatz (15), now with every level, $\phi_{ab} = \sum_{j=0}^4 g_j(\eta) \xi^j$, and expand the coefficients of (22) along the coordinate lines, $\alpha/\hat{m} = \sum_k A_k(\eta) \xi^k$, $\alpha \hat{m} = \sum_k B_k(\eta) \xi^k$ and $\hat{m} = \sum_k M_k(\eta) \xi^k$, $k \leq 4$. The chain (16) is kept as it stands, multiplied through by $\hat{m}$. With $S\phi_{ab}$ its right-hand side and $\psi_i = \sum_k B_k g'_{i-k}$ the coefficient of ξ^i in the normal flux, collecting ξ^i gives

$$(23) \quad \begin{aligned} g'_i &= \frac{1}{B_0} \left[\psi_i - \sum_{k \geq 1} B_k g'_{i-k} \right], \\ \psi'_i &= \alpha_e \sum_k M_k (Sg)_{i-k} - \sum_k A_k (i+2-k)(i+1) g_{i+2-k}, \end{aligned}$$

a missing level being zero. The first equation is triangular, so α is still never differentiated, and g_i and ψ_i are continuous through a jump on the coordinate line, level by level, so the march reads its pieces one-sidedly as before. On a flat line with α a function of η alone, (23) is (17) and the levels of the other parity stay zero. The initial data are those of Section 5.1; the parity trim and the shift identity do not survive.

Every seed to level four. The series coefficients are small: A_k and B_k are $O(h^k)$ and M_k is $O(h^{k+1})$ for $k \geq 1$. Level j of seed (a, b) is therefore $O(h^{j-a})$ above its top and contributes $O(h^{b+j})$ to a smooth solution, and an order count would keep it only while $b+j \leq 4$, 55 levels. The count hides a constant. Across a jump the far side of the seed of $\xi^a \eta^b$ carries the flux ratio $\alpha_e/\alpha_{\text{far}}$ to the power $\lceil b/2 \rceil$, and the dropped levels multiply it by the variation of α along the edge. On case 2 at $\delta = 0$ and 40,000 nodes the cut chain's error was 3.55×10^{-8} ; with every seed to level four it is 7.18×10^{-9} ,

and two more levels on every seed change nothing to two digits. The chain has 75 levels, 150 states.

Right-hand sides, series and scope. At the anchor only level zero of $L\phi$ survives, and its equation is marched with every term, so the chain holds on the whole line $\xi = 0$ and $(L\phi_{ab})(x_e)$ is $2\alpha_e$ on the two quadratics and zero otherwise. The entry $h_s\alpha_\xi$ of (19) is gone: the seed of ξ , now ξ plus lower levels driven by $A_1, A_2, \dots$, satisfies $L\phi_{10} = 0$, so the march carries the tangential derivative. The warp is ϕ_{01} with all its levels, and at the anchor $(LG)(x_e) = \alpha_e(G_{\xi\xi} + G_{\tilde{\eta}\tilde{\eta}}) + A_1(0)G_\xi$, the form of (21) with $A_1(0) = \partial_\xi(\alpha/\hat{m})$ in place of $h_s\alpha_\xi$. The coefficients A_k, B_k and M_k come from Fornberg's weights [For88] at $\xi = 0$ on 11 points 0.2 apart along the coordinate lines, whose positions are formed once per stencil and are exact. The foot edge's factor $\sigma(d/\delta)$ depends on η alone and is applied exactly; the other edge is taken as its pieces beyond 20δ of the sampled region and otherwise read through a 17-point Chebyshev interpolant of its distance along each sample line. Between 11 points at spacing 0.2 and 15 at 0.12 the weights of 40 rows of case 2 at 1250 nodes and $\delta = 0.0025$ move by 3.6×10^{-10} . At $\delta = 0$ the other curve may meet a stencil only as a coordinate line of the foot curve's frame, a parallel line or a concentric circle; no stencil of case 2 at 1250 nodes or more crosses both of its curves. A row costs 2.5 to 3.8 times a row of the frozen profile.

5.4. A thin resistive layer. The ring of case 3 is $1/s$ wide and thinner than the node spacing at every count (Section 2.3). No node lies inside it, and a stencil across its midline reaches both sides; its march crosses both circles in one pass along the radial normal, with both edges as segment boundaries. What the ring needs beyond the band is switched on only for a band whose two circles carry their gap, so that cases 1 and 2 are unchanged to the last bit.

The width and the series. At $s \geq 10^{10}$ the stored radii $0.35 - 1/s$ and 0.35 give the width only to 8×10^{-8} relative, and a march between two segment boundaries w/h_s apart, both of order one in stencil units, crosses the width only to $\text{ulp}(1)/(w/h_s)$, about 10^{-7} at $s = 10^{11}$. The band therefore carries the gap $1/s$ exactly, and the march runs in the offset from the outer circle's crossing, so that the segment it crosses is the gap to rounding. Across the ring $1/(\alpha\hat{m})$ is $1.5s$ at $\delta = 0$, and it multiplies the rounding of the sampled series into the small tangential levels, past the integrator's absolute tolerance; on a ring each segment's series are a piecewise Chebyshev interpolant in η , checked against the samples to 10^{-13} of their largest coefficient, so that the integrator's stages see a smooth function.

The smooth ring. With δ comparable to the width or larger, the band's composition (4) never reaches the ring's plateau and keeps only 0.41 of its contact resistance at $\delta = w/10$ and 0.01 at $w/4$ ($s = 10^3$); blending α by the difference of the two edges keeps a resistance of about w , against the jump's 1.5. Both lose the resistance that the jump ring holds fixed as s grows. The

smooth ring blends the resistivity instead,

$$(24) \quad \frac{1}{\alpha} = \frac{1}{\alpha_{\text{out}}} + \left(\frac{1}{\alpha_{\text{in}}} - \frac{1}{\alpha_{\text{out}}} \right) [\sigma(d_1/\delta) - \sigma(d_2/\delta)],$$

with d_1 and d_2 the signed distances to the inner and outer circles, so that its excess resistance $\int (1/\alpha - 1/\alpha_{\text{out}}) dn$ is the jump's, $w(1/\alpha_{\text{in}} - 1/\alpha_{\text{out}})$, at every (s, δ) . As $s \rightarrow \infty$ at fixed δ it tends to a layer of peak resistivity $R/(2\delta)$, with R the contact resistance, as the jump ring tends to the resistance itself. A resistive layer thinner than the grid is an imperfect interface in the limit [SP74, PP13, Has01], and strong-form schemes across such a contact need its jump conditions one derivative higher in two dimensions [LP03, LP06]; the flux seeds of Section 5.4 are that observation for the diffusion operator.

The flux seeds. With the fifteen seeds the rows beside the ring stall at fine counts (Section 6.6). Across a kink the flux enters the far side's values multiplied by the distance beyond it; across a thin resistive layer it enters with no such factor, as the jump $[u] = RF$ with F the flux through the layer. The seeds of degree four that carry a flux, those with $b \geq 1$, have a flux at the ring that is a polynomial of degree three in the offsets; the true flux differs from it by $O(h^4)$, which R turns into an $O(h^4)$ error in the far side's values and an $O(h^2)$ error in a row that reads many of them. Fitted by least squares to an exact mode through the ring, the fifteen seeds are $O(h^5)$ on a stencil's near side alone but $O(h^4)$ on the rows with 10 to 12 of their 30 nodes across it, where the jump-aware basis is $O(h^5)$. On a ring the block therefore adds the five seeds of degree five that carry a flux, those of $\xi^4\eta$, $\xi^3\eta^2$, $\xi^2\eta^3$, $\xi\eta^4$ and η^5 , from the chain of degree five: twenty seeds on thirty nodes, and with them the fit is $O(h^5)$. Their moment conditions vanish at the anchor, so the right-hand side is unchanged. An imperfect contact costs strong-form schemes a derivative in the jump conditions, as Lombard and Piraux observed for elastic waves [LP03, LP06]; the flux seeds add the functions of one degree higher that carry the flux, and nothing else. Why the jump-aware basis, whose continuity conditions also carry the flux only to degree three along the curve, fits at $O(h^5)$ with fifteen functions we have not established; its Cartesian frame couples the circle's curvature into the flux of its degree-four monomials, which is the likely source.

The Gaussians on a ring. Two changes to the Gaussians, both measured on the ring. The warp is the level zero of ϕ_{01} alone: its higher levels carry the ring's resistance, which varies 3 : 1 around it, along ξ into the far side's warped coordinate, a shear of about 4 per stencil radius at every h , and Gaussians on that coordinate put a floor under the ring's line, 3.3×10^{-6} at 80,000 nodes against 2.3×10^{-7} with level zero alone. Level zero has the full seed's flux and derivative at the anchor, so (21) is unchanged. And for an anchor inside a smooth ring, where α_e is a twentieth of the pieces' or less, the warp compresses the neighbours by that factor, so ε is taken from the nodes' spacing in the warped coordinates instead of the physical one.

Rows anchored in an edge. The warp's slope α_e/α is 1 across the anchor's side of the stencil only where the anchor sits on its piece. The resistivity tail of a smooth ring reaches several δ out, and an anchor in it has α_e below its piece's value, so the warp squeezes the anchor's own side, by 0.2 to 0.4 on the rows concerned. Their consistency is not the problem, since the truncation error on those rows keeps falling; their stability is: the anchor's share of such a row, $-w_e/\sum_{j \neq e} |w_j|$, is 0.08 to 0.13, against 0.25 to 0.27 with plain Gaussians. Plain Gaussians fail the other way where the edge is nearly resolved: some of those rows then get an anchor weight of the wrong sign. So on a ring a row whose anchor is off its piece, $|\ln(\alpha_e/\alpha_{\text{piece}})| > 0.05$ with α_{piece} the jump's value there, is solved with both Gaussian blocks on its one march and keeps the weights with the larger diagonal share, the diagonal dominance of [Mar16, App. B] with its sign; a row on its piece keeps the warp, and at $\delta = 0$ every anchor is on its piece. Off the ring the rule helps on the flat band where it fires and is mixed on case 2, where the row-by-row choice at near-ties can do worse than either uniform choice, so it stays the ring's safeguard.

6. TWO-DIMENSIONAL RESULTS

6.1. Test problems and references. [Stub, #49: the flat band, the sine pair, the ring; the separable, product-grid and reference-free references (stiff 4.1, 4.6, 4.8; statement 1).]

6.2. Naive baseline. [Stub, #49: the knee on scattered nodes and the jump construction on its floor (stiff 4.2; statements 2–3).]

6.3. Solvability and spectra. [Stub, #49: the seed rows in the global matrix (stiff 4.4; statement 6).]

6.4. The flat sweep. [Stub, #49: the seeds against the baselines at every (n, δ) , the warp, the elliptic ranking (stiff 4.5; statements 4, 5, 7).]

6.5. The curved feature. [Stub, #49: the frozen profile and the tangential chain on the sine pair (stiff 4.6–4.7; statement 8).]

6.6. The ring and the extremizing sweep. [Stub, #49: the twins of EABE Fig. 19–20 and the smooth ring (stiff 4.8, 4.10; statement 9).]

6.7. Coefficient treatments. [Stub, #49: the treatments on scattered nodes (stiff 4.9; statement 10).]

6.8. A snapshot. [Stub, #49: the four regimes on one node set (stiff 5.1; statement 11).]

7. DISCUSSION

7.1. Limitations. [Stub, #50: corners, extreme contrasts, the cost in three dimensions, what the experiments leave open.]

7.2. Future work. *[Stub, #50: in the order to do it.]*

7.3. Conclusions. *[Stub, #50: LITERATURE.md section 6b, verbatim.]*

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[Stub, #52: acknowledgments; where the code and data live.]

TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

[Stub, #52.]

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